comarison metabarcoding and microscopy

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Comarison metabarcoding and microscopy

library

install

Activate

```r
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(ggplot2)
library(tibble)
library(vegan)
```

```
Loading required package: permute
```

```r
library(knitr)
library(lubridate)
```

```
Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```r
library(stringr)
library(forcats)
library(purrr)
library(patchwork)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v readr 2.1.5

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```r
library(broom)
```

Retrieve data

Home pc

```r
# microscopy

R_5_abundance_2023 <- read_excel(
  "C:/Program Files (x86)/Results/Planteplanktontellinger/R-5 2023/R-5_Ringdalsfjorden_kar
  na = "."
)
```

```
New names:
* `` -> `...1`
```

```r
R_5_abundance_2024 <- read_excel(
  "C:/Program Files (x86)/Results/Planteplanktontellinger/R-5 2024/R-5_Ringdalsfjorden_kar
  na = "."
)
```

```
New names:
* `` -> `...1`
```

```r
S_9_abundance_2023 <- read_excel(
  "C:/Program Files (x86)/Results/Planteplanktontellinger/S-9 2023/S-9_Haslau_karbon_2023.
 na = "."
)
```

```
New names:
* `` -> `...1`
```

```r
S_9_abundance_2024 <- read_excel(
  "C:/Program Files (x86)/Results/Planteplanktontellinger/S-9 2024/S-9_Haslau_karbon_2024.
  na = "."

)
```

```
New names:
* `` -> `...1`
```

```r
#metabarcoding
setwd("C:/Results")

metadata <- read_excel("samples Oslo Fjord.xlsx")



Metabarcoding_results <- read_excel("Oslofjord_illumina_dada2.xlsx")
```

school pc

data cleaning

microscopy

```r
clean_abundance_data <- function(df) {
  df %>%
    # rename first two columns
    rename(higher_taxa = 1, Species = 2) %>%

    # remove rows where both columns are empty
    filter(!(is.na(higher_taxa) & is.na(Species))) %>%

    # fill down higher_taxa names
    fill(higher_taxa, .direction = "down") %>%

    # remove rows without species
    filter(!is.na(Species) & Species != "") %>%

    # remove summary rows
    filter(!(Species %in% c("Sum:", "Sum totalt:"))) %>%

    # convert "." to NA if any remain (for safety)
    mutate(across(everything(), ~na_if(., "."))) %>%

    # convert numeric columns to numeric if read as character
    mutate(across(-c(higher_taxa, Species), as.numeric)) %>%

    # convert abundance values to relative abundance (%)
    mutate(across(-c(higher_taxa, Species),
                  ~ (. / sum(., na.rm = TRUE)) * 100))
}

# Apply to your datasets
R_5_abundance_2023_clean <- clean_abundance_data(R_5_abundance_2023)
R_5_abundance_2024_clean <- clean_abundance_data(R_5_abundance_2024)
S_9_abundance_2023_clean <- clean_abundance_data(S_9_abundance_2023)
S_9_abundance_2024_clean <- clean_abundance_data(S_9_abundance_2024)

TAXON_COLORS_micro <- c(
```

```
  "Chlorophyta"    = "#1f78b7",
  "Ciliophora"     = "#ff7f00",
  "Cryptophyta"    = "#6a3d9a",
  "Diatoms"        = "blue",
  "Dinophyta" = "#01a01A",
  "Euglenophyta" = "#100",
  "Haptophyta"     = "#a6cee3",
  "Other Haptophytes" ="#ee87ee",
  "Other Ochrophyta" = "#e31a1c",
  "Other Protists" = "#656739",
  "Stramenopiles"  = "#b15928"

)



# removing unnusable letters
# data frame names to process
df_names <- c(
  "R_5_abundance_2023_clean",
  "R_5_abundance_2024_clean",
  "S_9_abundance_2023_clean",
  "S_9_abundance_2024_clean"
)

# target keywords and their canonical replacements
replacements <- c(
  chlorophyta      = "Chlorophyta",
  euglenophyceae   = "Euglenophyceae",
  cyanobacteria    = "Cyanobacteria",
  prasinophyceae   = "Prasinophyceae"
)

for (nm in df_names) {
  if (!exists(nm, envir = .GlobalEnv)) {
    warning("Object '", nm, "' not found; skipping.")
    next
  }
  df <- get(nm, envir = .GlobalEnv)
  if (!"higher_taxa" %in% names(df)) {
    warning("Data frame '", nm, "' has no column 'higher_taxa'; skipping.")
    next
```

```r
  }

  # ensure character and trim
  df$higher_taxa <- trimws(as.character(df$higher_taxa))

  # for each keyword, find rows containing it and set to canonical form
  for (kw in names(replacements)) {
    mask <- grepl(kw, df$higher_taxa, ignore.case = TRUE, perl = TRUE)
    if (any(mask)) df$higher_taxa[mask] <- replacements[[kw]]
  }

  # tidy whitespace
  df$higher_taxa <- gsub("\\s+", " ", df$higher_taxa)
  df$higher_taxa <- trimws(df$higher_taxa)

  assign(nm, df, envir = .GlobalEnv)
  message(sprintf("Processed %s: replaced %d rows matching targets.",
                  nm, sum(grepl(paste(names(replacements), collapse="|"), df$higher_taxa,
}
```

```
Processed R_5_abundance_2023_clean: replaced 21 rows matching targets.

Processed R_5_abundance_2024_clean: replaced 14 rows matching targets.

Processed S_9_abundance_2023_clean: replaced 11 rows matching targets.

Processed S_9_abundance_2024_clean: replaced 13 rows matching targets.
```

```r
# named mapping: names = old values, values = new values
mapping <- c(
  "Bacillariophyceae (kiselalger)" = "Diatoms",
  "Chlorophyta" = "Chlorophyta",
  "Choanoflagellatea (krageflagellater)" = "Other Protists",
  "Chrysophyceae (gullalger)" = "Other Ochrophyta",
  "Ciliophora (ciliater)" = "Ciliophora",
  "Coccolithophyceae (kalk- og svepeflagellater)" = "Haptophyta",
  "Conjugatophyceae" = "Chlorophyta",
  "Cryptophyceae (svelgflagellater)" = "Cryptophyta",
  "Dictyochophyceae (kiselflagellater og pedineller)" = "Other Ochrophyta",
  "Dinophyceae (fureflagellater)" = "Dinophyta",
```

```r
    "Ebriophyceae (skjelettflagellater)" = "Other Protists",
    "Euglenophyceae" = "Euglenophyta",
    "Imbricatea" = "Other Protists",
    "Prasinophyceae" = "Chlorophyta",
    "Pyramimonadophyceae" = "Chlorophyta",
    "Synurophyceae" = "Other Ochrophyta",
    "Telonemea" = "Other Protists"
  )

  # rows to drop
  drop_these <- c(
    "Cyanobacteria",
    "Classes incertae sedis (ubestemte klasser)"
  )

  # put your datasets in a named list
  dfs <- list(
    R_5_abundance_2023_clean = R_5_abundance_2023_clean,
    R_5_abundance_2024_clean = R_5_abundance_2024_clean,
    S_9_abundance_2023_clean = S_9_abundance_2023_clean,
    S_9_abundance_2024_clean = S_9_abundance_2024_clean
  )

  # function to remap and filter one dataframe
  fix_taxa <- function(df) {
    df %>%
      # make sure higher_taxa is character to map reliably
      mutate(higher_taxa = as.character(higher_taxa)) %>%
      # replace only exact matches listed in mapping
      mutate(higher_taxa = dplyr::recode(higher_taxa, !!!mapping)) %>%
      # remove unwanted groups
      filter(!higher_taxa %in% drop_these)
  }

  # apply to all and reassign back to the global env (or keep as a list)
  fixed_list <- lapply(dfs, fix_taxa)

  # if you want to overwrite the original objects in the global environment:
  list2env(fixed_list, envir = .GlobalEnv)
```

```
<environment: R_GlobalEnv>
```

```
#fix relative abundance after the removal of certain groups
# convert abundance values to relative abundance (%)
relative_abundance <- function(df) {
  df %>%
mutate(across(-c(higher_taxa, Species),
                  ~ (. / sum(., na.rm = TRUE)) * 100))}

R_5_abundance_2023_clean <- relative_abundance(R_5_abundance_2023_clean)
R_5_abundance_2024_clean <- relative_abundance(R_5_abundance_2024_clean)
S_9_abundance_2023_clean <- relative_abundance(S_9_abundance_2023_clean)
S_9_abundance_2024_clean <- relative_abundance(S_9_abundance_2024_clean)
```

Metabarcoding

```
# Keep only rows with the station IDs i need and all negatives
metadata <- subset(metadata, station_id %in% c("R-5", "S-9", "Negative"))

# remove the negatives i dont need
metadata <- subset(metadata, !(file_name_R1 %in% c("14-OS24_S14_L001_R1_001.fastq.gz", "21


cols_to_remove <- c(
  "1-OS51", "2-OS52", "3-OS57", "4-OS23", "5-OS46", "6-OS30", "7-OS50",
  "8-OS17", "9-OS18", "10-OS19", "11-OS20", "12-OS21", "13-OS22", "14-OS24",
  "15-OS25", "16-OS26", "17-OS27", "18-OS28", "19-OS29", "20-OS31", "21-OS32",
  "22-OS41", "23-OS43", "24-OS44", "25-OS45", "26-OS47", "27-OS48", "28-OS49",
  "29-OS53", "30-OS54", "31-OS55", "32-OS56", "33-OS42", "34-OS44_2",
  "35-OS45_2", "36-OS47_2", "140-S28", "141-S29"
)

Metabarcoding_results <- Metabarcoding_results %>%
  select(-any_of(cols_to_remove))



# remove unneccessary samples (species i dont look at)
filtered_results <- Metabarcoding_results %>%
  filter(
    division %in% c("Stramenopiles", "Chlorophyta", "Cryptophyta", "Haptophyta", "Telonemi
    subdivision %in% c("Dinoflagellata", "Ciliophora") |
```

```r
    class %in% c("Bacillariophyceae", "Coscinodiscophyceae", "Mediophyceae",
                 "Chrysophyceae", "Dictyochophyceae", "Choanoflagellatea", "Ebriophyceae")
  )

# helper vectors
division_wanted    <- c("Stramenopiles", "Chlorophyta", "Cryptophyta", "Haptophyta", "Telo
subdivision_wanted <- c("Dinoflagellata", "Ciliophora")
class_wanted       <- c("Bacillariophyceae", "Coscinodiscophyceae", "Mediophyceae",
                        "Chrysophyceae", "Dictyochophyceae", "Choanoflagellatea", "Ebrioph



station_cols <- c("113-S1","114-S2","115-S3","116-S4","117-S5","118-S6",
                  "119-S7","120-S8","121-S9","122-S10","123-S11","124-S12",
                  "125-S13","126-S14","127-S15","128-S16","129-S17","130-S18",
                  "131-S19","132-S20","133-S21","134-S22","135-S23","136-S24",
                  "137-S25","138-S26","139-S27","142-S30",
                  "143-S31","144-S32")


# Combined legend order BEFORE merging
all_taxa <- c(class_wanted, subdivision_wanted, division_wanted)


# remove samples with a bootstrap value bellow 80 at subdivision
filtered_results <- filtered_results[filtered_results$subdivision_boot >= 80, ]

#change from reads to %
cols <- c("113-S1","114-S2","115-S3","116-S4","117-S5","118-S6","119-S7","120-S8","121-S9"

# Convert each column to percentage of its total
filtered_results[cols] <- lapply(
  filtered_results[cols],
  function(x) (x / sum(x, na.rm = TRUE)) * 100
)

# Filter ASVs: keep only rows with mean relative abundance >= 0.001
filtered_results_clean <- filtered_results[
  rowMeans(filtered_results[, cols]) >= .001,
]
# Convert each column to percentage of its total
```

```r
filtered_results_clean[cols] <- lapply(
  filtered_results_clean[cols],
  function(x) (x / sum(x, na.rm = TRUE)) * 100)

TAXON_COLORS_meta <- c(
  "Diatoms"        = "blue",
  "Dinophyta" = "#01a01A",
  "Ciliophora"     = "#ff7f00",
  "Stramenopiles"  = "#b15928",
  "Chlorophyta"    = "#1f78b7",
  "Cryptophyta"    = "#6a3d9a",
  "Haptophyta"     = "#a6cee3",
    "Euglenophyta" = "#100",
  "Other Ochrophyta" = "#e31a1c",
  "Other Protists" = "#656739"


  )

#long format
data_long_meta <- filtered_results_clean %>%
  mutate(taxon = case_when(
    !is.na(class) & class %in% class_wanted ~ class,
    !is.na(subdivision) & subdivision %in% subdivision_wanted ~ subdivision,
    !is.na(division) & division %in% division_wanted ~ division,
    TRUE ~ NA_character_
  )) %>%
  filter(!is.na(taxon)) %>%
  select(taxon, all_of(station_cols)) %>%
  pivot_longer(
    cols = all_of(station_cols),
    names_to = "station",
    values_to = "value"
  ) %>%
  mutate(value = as.numeric(replace_na(value, 0)))

data_long_meta <- data_long_meta %>%
  mutate(taxon = case_when(
    taxon %in% c("Bacillariophyceae", "Mediophyceae", "Coscinodiscophyceae") ~ "Diatoms",
    TRUE ~ taxon
  ))
```

```r
plot_df <- data_long_meta %>%
  group_by(station, taxon) %>%
  summarise(total = sum(value, na.rm = TRUE), .groups = "drop") %>%
  tidyr::complete(station, taxon = all_taxa, fill = list(total = 0)) %>%
  group_by(station) %>%
  mutate(percentage = if (sum(total) == 0) 0 else 100 * total / sum(total)) %>%
  ungroup()

plot_df <- plot_df %>%
  mutate(taxon = case_when(
    taxon %in% c("Bacillariophyceae", "Mediophyceae", "Coscinodiscophyceae") ~ "Diatoms",
    TRUE ~ taxon
  ))

all_taxa_merged <- c(
  "Diatoms",
  "Dinophyta",
  "Ciliophora",
  "Other Protists",
  "Chlorophyta",
  "Cryptophyta",
  "Haptophyta",
  "Euglenophyta",
  "Other Ochrophyta"

)

seasonal_input <- plot_df %>%
  select(station, taxon, percentage)

#rename so names of the differing data sets match
data_long_meta <- data_long_meta %>% rename(sample_id = station)

seasonal_input <- seasonal_input %>%
  rename(sample_id = station)

# Clean file_name_R1 in metadata
metadata <- metadata %>%
  mutate(sample_id = gsub("(_.*)$", "", file_name_R1))

#combine
```

```r
combined <- seasonal_input %>%
  left_join(metadata, by = "sample_id")

combined_filtered <- combined %>%
  filter(station_id != "Negative")

# Merge diatom classes into "Diatoms"
combined_filtered <- combined_filtered %>%
  mutate(
    taxon = case_when(
      taxon %in% c("Bacillariophyceae", "Mediophyceae", "Coscinodiscophyceae") ~ "Diatoms"
      TRUE ~ taxon
    )
  )



# renaming to match microscopy
mapping <- c(
  "Choanoflagellatea" = "Other Protists",
  "Chrysophyceae"     = "Other Ochrophyta",
  "Dictyochophyceae"  = "Other Ochrophyta",
  "Ebriophyceae"      = "Other Protists",
  "Filosa-Imbricatea" = "Other Protists",
  "Telonemia"        = "Other Protists",
  "Stramenopiles"     = "Other Ochrophyta",
  "Dinoflagellata" = "Dinophyta"

)

# columns to apply mapping to (change as needed)
cols_to_fix <- c("taxon")

combined_filtered <- combined_filtered %>%
  mutate(across(all_of(cols_to_fix), ~ {
    # work with character to avoid factor problems
    x <- as.character(.x)
    # replace where there is a match in mapping
    x[x %in% names(mapping)] <- mapping[x[x %in% names(mapping)]]
    # optionally convert back to factor (comment out if you want character)
    # factor(x)
```

```
    x
  }))


# Make sure factor levels match your color palette
combined_filtered <- combined_filtered %>%
  mutate(taxon = factor(taxon, levels = all_taxa_merged))

#remove unecessary columns
data_long_meta <- combined_filtered %>%
  select(any_of(c("taxon", "percentage", "station_id", "date")))

#rename columns to match microscopy columns
data_long_meta <- data_long_meta %>%
  rename(
    higher_taxa = taxon,
    station     = station_id,   # change to station_id if your column is named station_id
    Abundance   = percentage,
    Date        = date
  )
```

make datasett to show abundance at different dates

```
data_agg <- data_long_meta %>%
  group_by(higher_taxa, station, Date) %>%
  summarize(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop")


# S-9 (allow S-9, S_9, S9, s-9, etc.)
data_S9 <- data_agg %>%
  filter(grepl("S[-_]?9", station, ignore.case = TRUE)) %>%
  select(-station)

# R-5 (allow R-5, R_5, R5, r-5, etc.)
data_R5 <- data_agg %>%
  filter(grepl("R[-_]?5", station, ignore.case = TRUE)) %>%
  select(-station)

# Optional: see rows that did not match either pattern
unmatched <- data_agg %>%
  filter(!grepl("S[-_]?9", station, ignore.case = TRUE) &
```

```r
        !grepl("R[-_]?5", station, ignore.case = TRUE))

if (nrow(unmatched) > 0) {
  message("There are ", nrow(unmatched), " rows with stations that are neither S-9 nor R-5
}

data_wide_R5 <- data_R5 %>%
  # ensure Date is character for clean column names (optional)
  mutate(Date = as.character(Date)) %>%
  # if abundance might be non-numeric
  mutate(abundance = as.numeric(Abundance)) %>%
  # pivot, summing duplicates and filling missing with 0
  pivot_wider(
    names_from  = Date,
    values_from = abundance,
    values_fill = 0,
    values_fn   = sum
  )

data_wide_S9 <- data_S9 %>%
  # ensure Date is character for clean column names (optional)
  mutate(Date = as.character(Date)) %>%
  # if abundance might be non-numeric
  mutate(abundance = as.numeric(Abundance)) %>%
  # pivot, summing duplicates and filling missing with 0
  pivot_wider(
    names_from  = Date,
    values_from = abundance,
    values_fill = 0,
    values_fn   = sum
  )


collapse_same_taxa <- function(df) {
  # ensure input is a data.frame/tibble
  if (!is.data.frame(df)) stop("Input must be a data.frame/tibble")

  # detect taxa column (you already have higher_taxa)
  taxa_col <- "higher_taxa"

  # columns to sum: everything except the taxa column
```

```
  value_cols <- setdiff(names(df), taxa_col)
  if (length(value_cols) == 0) stop("No value/date columns found to sum")

  df %>%
    # coerce all value columns to numeric (handles factors/characters)
    mutate(across(all_of(value_cols), ~ as.numeric(as.character(.)))) %>%
    group_by(across(all_of(taxa_col))) %>%
    summarize(across(all_of(value_cols), ~ sum(.x, na.rm = TRUE)), .groups = "drop")
}

# Apply to your datasets
data_wide_R5_collapsed <- collapse_same_taxa(data_wide_R5)
data_wide_S9_collapsed <- collapse_same_taxa(data_wide_S9)

library(dplyr)

wide_R5 <- data_wide_R5_collapsed %>% select(-one_of("Abundance"))
wide_S9 <- data_wide_S9_collapsed %>% select(-one_of("Abundance"))
```

plots

microscopy

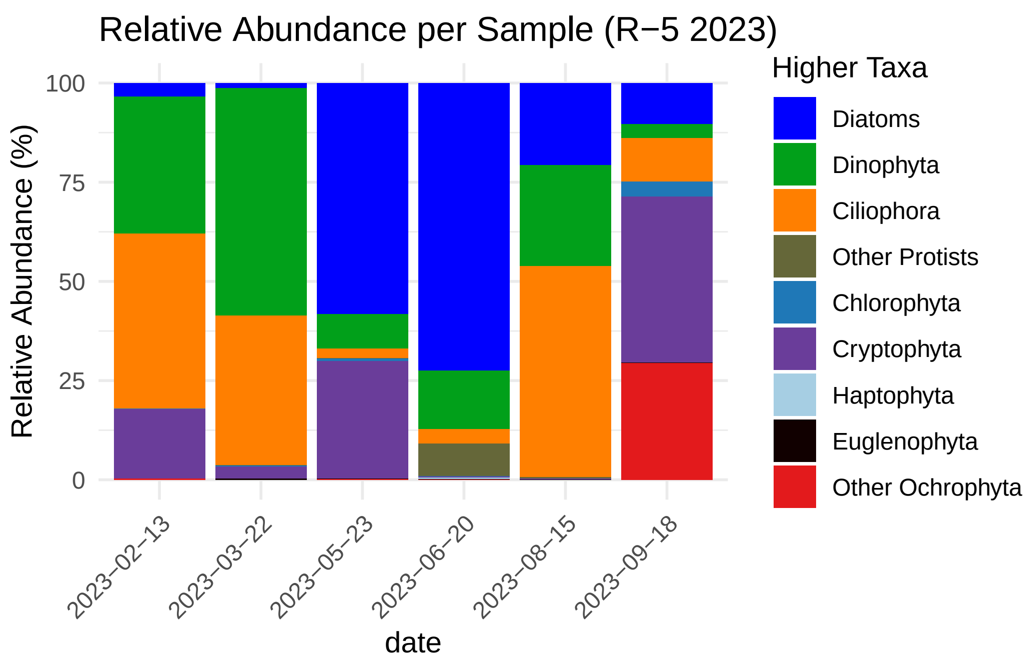
```
all_taxa_merged_micro <- c(
  "Diatoms",
  "Dinophyta",
  "Ciliophora",
  "Other Protists",
  "Chlorophyta",
  "Cryptophyta",
  "Haptophyta",
  "Euglenophyta",
  "Other Ochrophyta"
  )

# List of datasets to process (named so titles/filenames are easy)
datasets <- list(
  "R-5 2023" = R_5_abundance_2023_clean,
  "R-5 2024" = R_5_abundance_2024_clean,
```

```
  "S-9 2023" = S_9_abundance_2023_clean,
  "S-9 2024" = S_9_abundance_2024_clean
)

# Helper function: convert one dataset to long and set factor levels
make_long <- function(df) {
  df_long <- df %>%
    pivot_longer(
      cols = -c(higher_taxa, Species),
      names_to = "Sample",
      values_to = "Abundance"
    ) %>%
    mutate(
      Sample = dmy(Sample)      # convert Sample from string to Date
    ) %>%
    arrange(Sample) %>%
    mutate(
      Sample = factor(Sample),  # convert back to factor ordered by date
      higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
    )
  df_long
}

# Convert all datasets to long format
long_list <- map(datasets, make_long)

# Helper to create a plot for one long dataframe and a title
make_plot <- function(df_long, title_text) {
  ggplot(df_long, aes(x = Sample, y = Abundance, fill = higher_taxa)) +
    geom_bar(stat = "identity", position = "stack") +
    scale_fill_manual(values = TAXON_COLORS_micro, na.value = "red") +
    labs(
      title = paste("Relative Abundance per Sample (", title_text, ")", sep = ""),
      x = "date",
      y = "Relative Abundance (%)",
      fill = "Higher Taxa"
    ) +
    theme_minimal() +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "right"
```

```r
    )
}

# Create plots for each long dataset
plot_list <- map2(long_list, names(long_list), make_plot)

list(plots = plot_list)
```

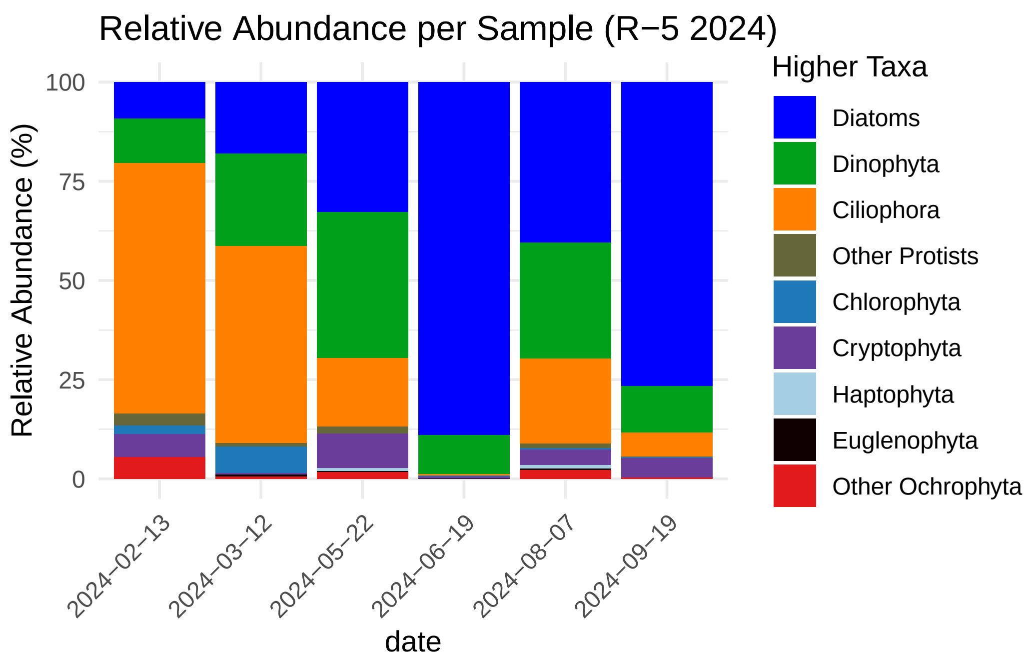
```
$plots
$plots$`R-5 2023`
```

```
Warning: Removed 459 rows containing missing values or values outside the scale range
(`geom_bar()`).
```

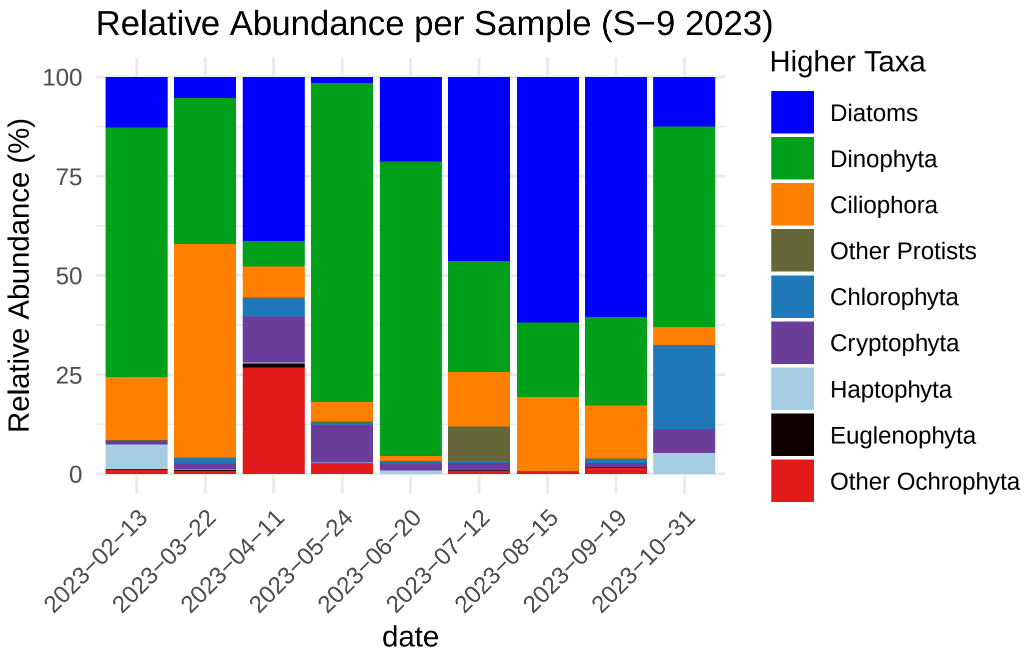



```
$plots$`R-5 2024`
```

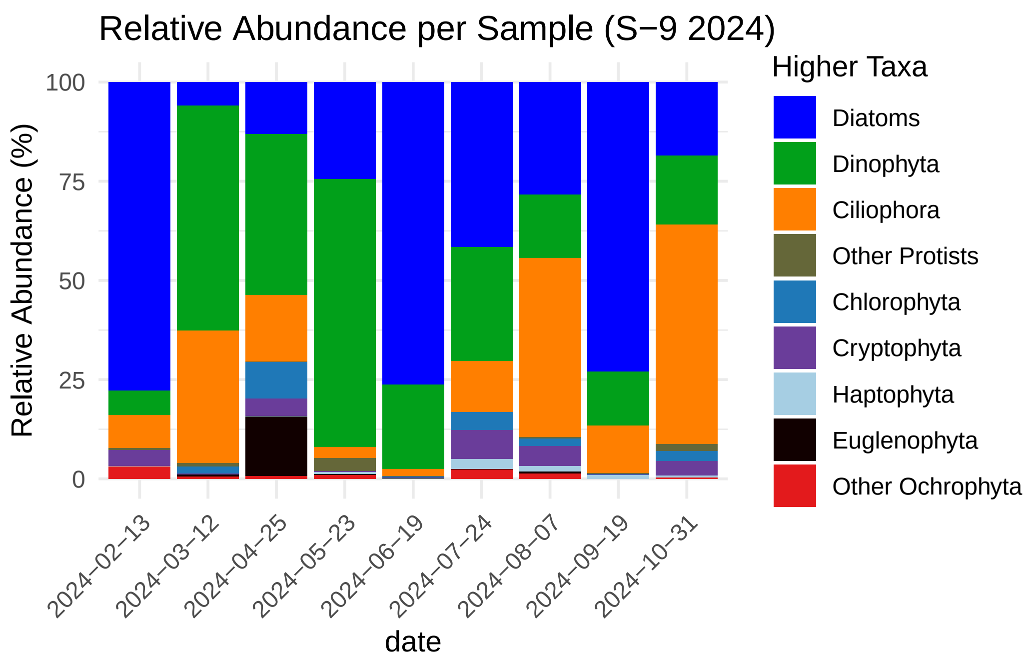
```
Warning: Removed 496 rows containing missing values or values outside the scale range
(`geom_bar()`).
```




```
$plots$`S-9 2023`

Warning: Removed 852 rows containing missing values or values outside the scale range
(`geom_bar()`).
```



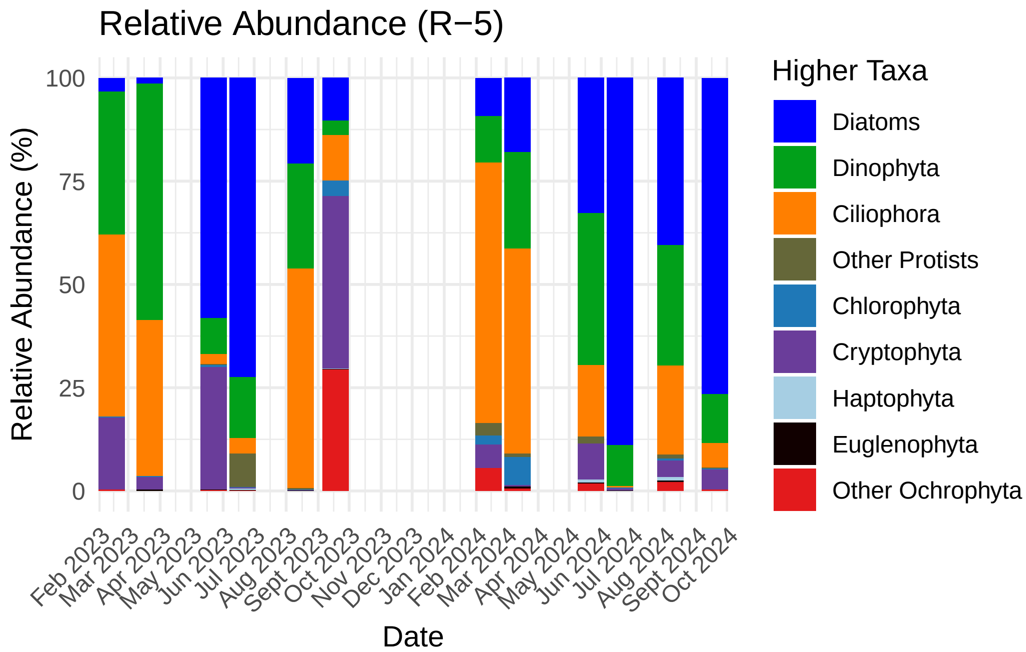

```
$plots$`S-9 2024`

Warning: Removed 1053 rows containing missing values or values outside the scale range
(`geom_bar()`).
```




```
#adding station S-9 or R-5 to the datasets
long_list_with_station <- imap(long_list, ~ .x %>%
  mutate(station = if (grepl("^S-9", .y)) "S-9" else "R-5")
)

# combining microscopy datasets into one dataset
combined_long_microscopy <- bind_rows(long_list_with_station, .id = "dataset")

#removing column dataset
combined_long_microscopy <- combined_long_microscopy %>%
  select(-dataset)

#rename sample column to date
combined_long_microscopy <- combined_long_microscopy %>%
  rename(Date = Sample)

#1) Robustly convert Date and prepare data
combined_long_microscopy <- combined_long_microscopy %>%
mutate(
Date = as.character(Date),     # handle factor/character/POSIXt uniformly
Date = ymd(Date),              # parse "yyyy-mm-dd"
Abundance = as.numeric(Abundance),
```

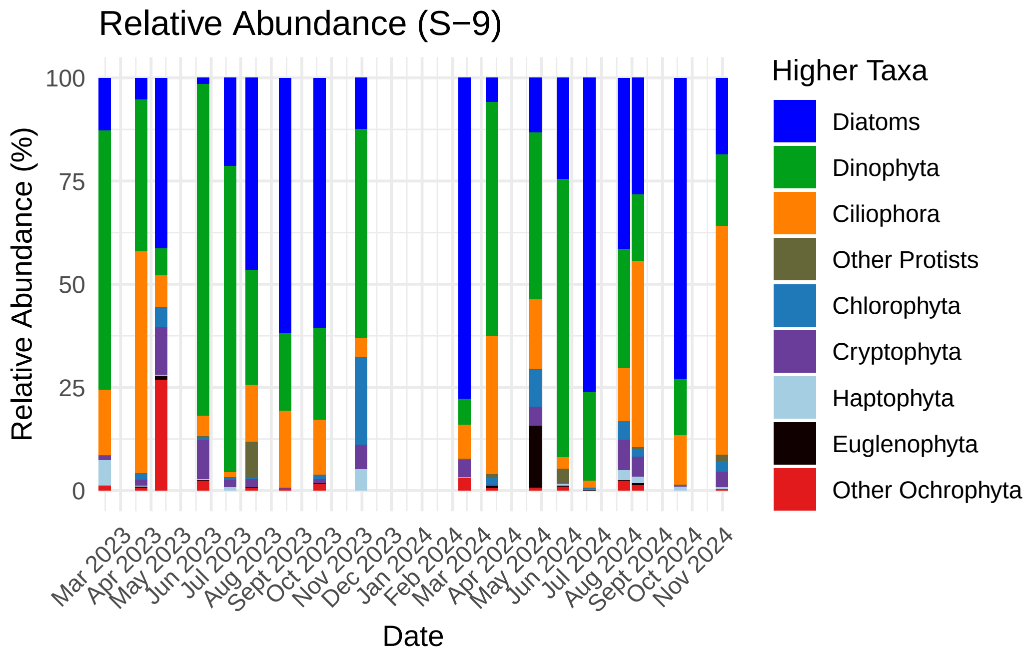
```r
higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
)
# Check for failed parses (optional)
if (any(is.na(combined_long_microscopy$Date))) {
warning("Some Date rows failed to parse. Inspect with:\n",
"combined_long_microscopy %>% filter(is.na(Date)) %>% distinct(Date)")
}
# 2) Filter to 2023-01-01 through 2024-12-31
date_min <- as.Date("2023-01-01")
date_max <- as.Date("2024-12-31")
df_2023_2024 <- combined_long_microscopy %>%
filter(Date >= date_min & Date <= date_max)


# 3) Create plots --- one for R-5 and one for S-9
make_station_plot <- function(df, station_name) {
df_station <- df %>% filter(station == station_name)
ggplot(df_station, aes(x = Date, y = Abundance, fill = higher_taxa)) +
geom_col(position = "stack") +
scale_fill_manual(values = TAXON_COLORS_micro, na.value = "red") +
scale_x_date(date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
labs(
title = paste0("Relative Abundance (", station_name, ")"),
x = "Date",
y = "Relative Abundance (%)",
fill = "Higher Taxa"
) +
theme_minimal() +
theme(
axis.text.x = element_text(angle = 45, hjust = 1),
legend.position = "right"
)
}
p_R5 <- make_station_plot(df_2023_2024, "R-5")
p_S9 <- make_station_plot(df_2023_2024, "S-9")
# Print plots
print(p_R5)
```

```
Warning: Removed 955 rows containing missing values or values outside the scale range
(`geom_col()`).
```



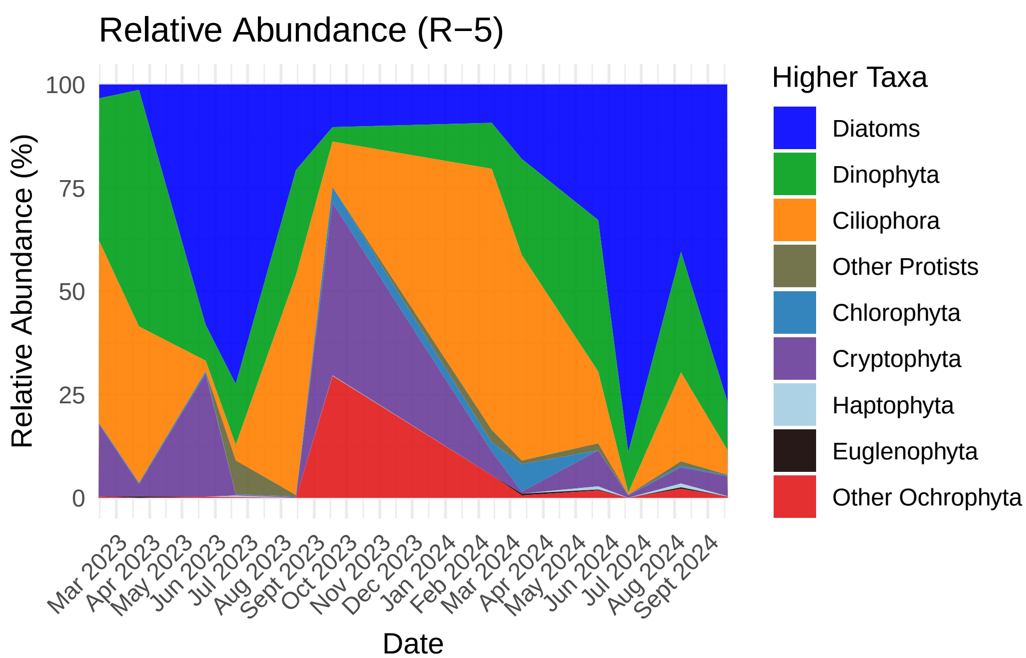

```
print(p_S9)
```

```
Warning: Removed 1905 rows containing missing values or values outside the scale range
(`geom_col()`).
```

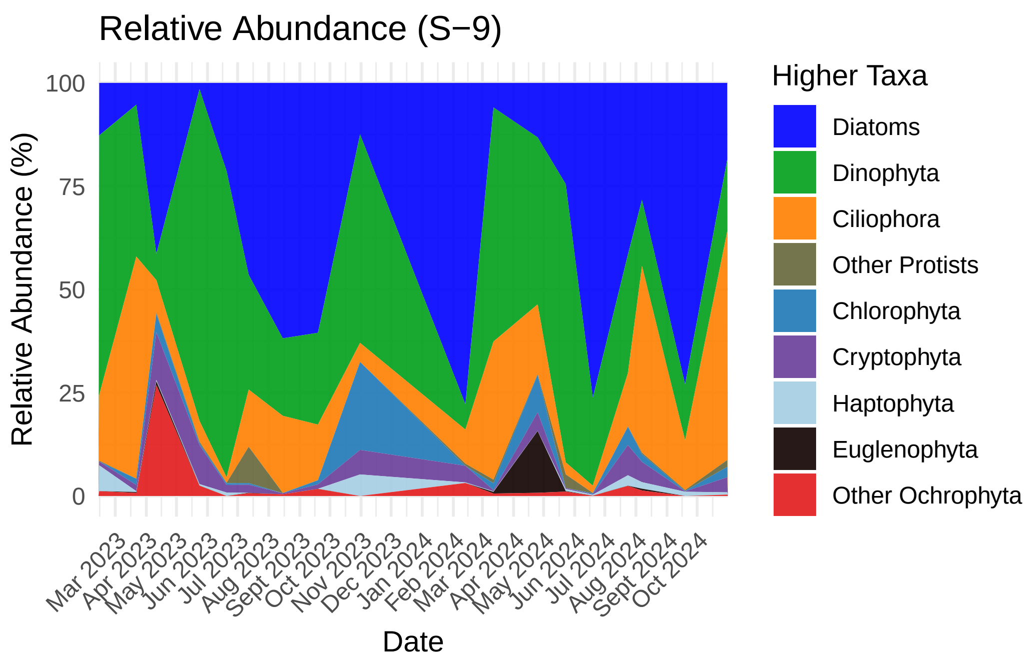



```
# Prepare data: ensure types, restrict to 2023-2024, and aggregate by Date/station/higher_
df_prep <- combined_long_microscopy %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
# parse "yyyy-mm-dd"
higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
) %>%
filter(!is.na(Date)) %>%
filter(Date >= as.Date("2023-01-01") & Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
  summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop")

make_area_plot <- function(df, station_name) {
df_station <- df %>% filter(station == station_name)
ggplot(df_station, aes(x = Date, y = Abundance, fill = higher_taxa)) +
geom_area(position = "stack", color = NA, alpha = 0.9) +
scale_fill_manual(values = TAXON_COLORS_micro, na.value = "red") +
scale_x_date(date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
labs(
```

```r
title = paste0("Relative Abundance (", station_name, ")"),
x = "Date",
y = "Relative Abundance (%)", # adjust if Abundance is proportions
fill = "Higher Taxa"
) +
theme_minimal() +
theme(
axis.text.x = element_text(angle = 45, hjust = 1),
legend.position = "right"
)
}
# Create area plots
area_R5 <- make_area_plot(df_prep, "R-5")
area_S9 <- make_area_plot(df_prep, "S-9")
# Print plots
print(area_R5)
```




```r
print(area_S9)
```

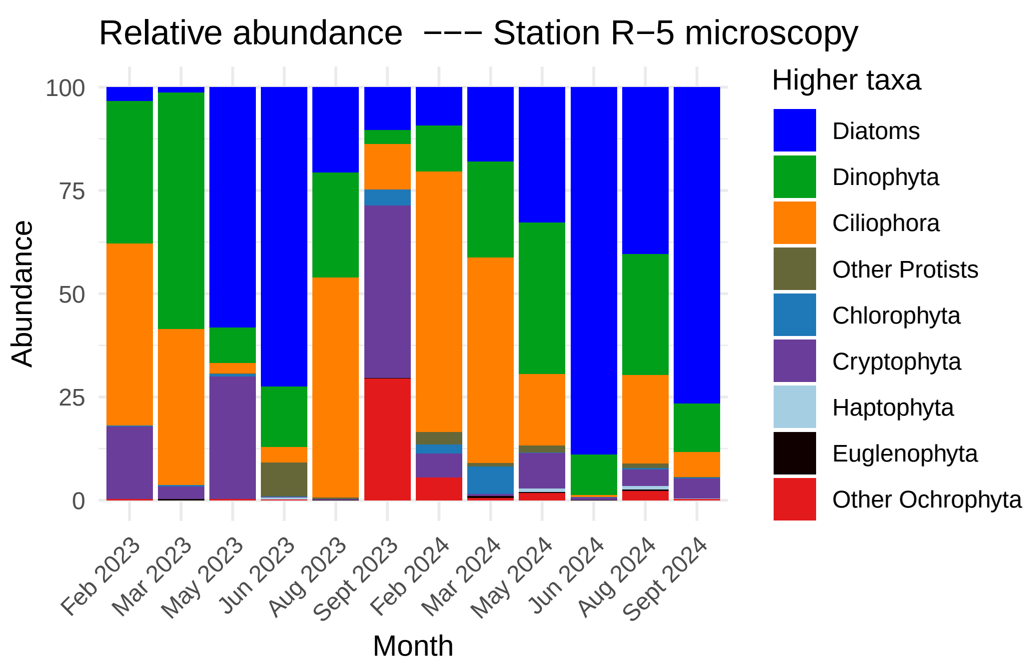


new plots

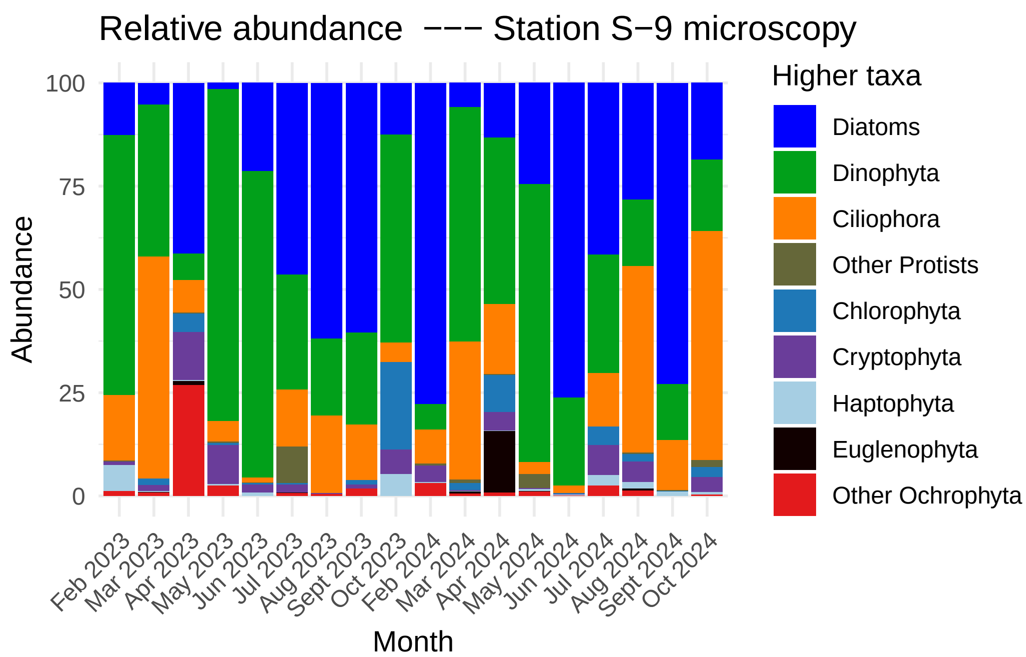
```
# Robust parsing orders - add formats you know your data uses if needed
orders <- c("ymd","dmy","mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM", "dmY", "Y/m/d", "d/m

# Recreate df_plot from df_2023_2024
df_plot <- df_2023_2024 %>%
  # remove rows with missing Abundance
  filter(!is.na(Abundance)) %>%
  mutate(
    raw_Date = as.character(Date),
    # try parsing with lubridate (returns POSIXct); suppress warnings so pipeline doesn't
    parsed_dt = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)
    parsed_Date = as_date(parsed_dt),      # convert to Date (or NA if parsing failed)
    # build a date_label fallback (we'll create month_label next)
    date_label = if_else(!is.na(parsed_Date),
                         format(parsed_Date, "%Y-%m-%d"),
                         raw_Date),
    date_order_key = if_else(!is.na(parsed_Date), as.numeric(parsed_Date), NA_real_)
  )
```

```
# Create month_label like "Feb 2023" (use parsed_Date when available, else use raw text)
df_plot <- df_plot %>%
  mutate(
    month_label = if_else(!is.na(parsed_Date),
                          format(parsed_Date, "%b %Y"),
                          raw_Date),
    # keep month label as capitalized month (e.g., "Feb 2023")
    month_order_key = if_else(!is.na(parsed_Date), as.numeric(parsed_Date), NA_real_)
  )

# Build ordered factor levels so months are chronological (parsed ones first), then any un
parsed_levels <- df_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
  pull(month_label) %>%
  unique()

unparsed_levels <- df_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_plot <- df_plot %>%
  mutate(month_label = factor(month_label, levels = all_levels, ordered = TRUE))

# Helper to make plot for a station
make_station_bar_month <- function(station_name, color_map = NULL) {
  df_plot %>%
    filter(station == station_name) %>%
    group_by(month_label, higher_taxa) %>%
    summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
    ggplot(aes(x = month_label, y = Abundance, fill = higher_taxa)) +
    geom_col(position = "stack", width = 0.9) +
    { if (!is.null(color_map)) scale_fill_manual(values = color_map, na.value = "grey50")
    labs(title = paste0("Relative abundance  --- Station ", station_name, " microscopy"),
         x = "Month", y = "Abundance", fill = "Higher taxa") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}
```

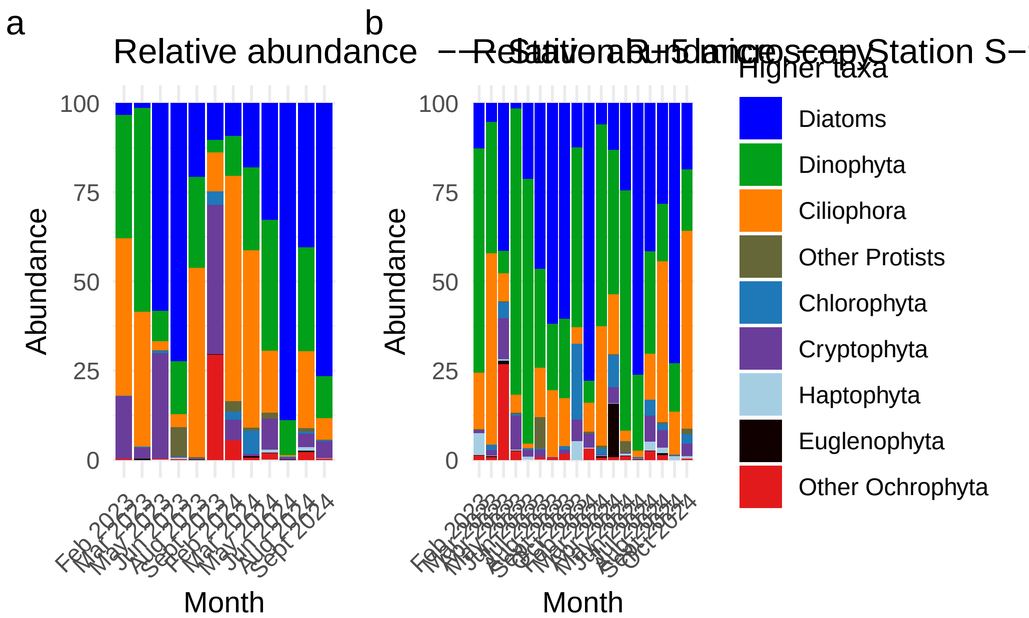
```r
# Create plots (use TAXON_COLORS_micro if available; otherwise pass NULL)
if (exists("TAXON_COLORS_micro")) {
  p_R5 <- make_station_bar_month("R-5", TAXON_COLORS_micro)
  p_S9 <- make_station_bar_month("S-9", TAXON_COLORS_micro)
} else {
  p_R5 <- make_station_bar_month("R-5", NULL)
  p_S9 <- make_station_bar_month("S-9", NULL)
}

# Print plots
print(p_R5)
```




```r
print(p_S9)
```



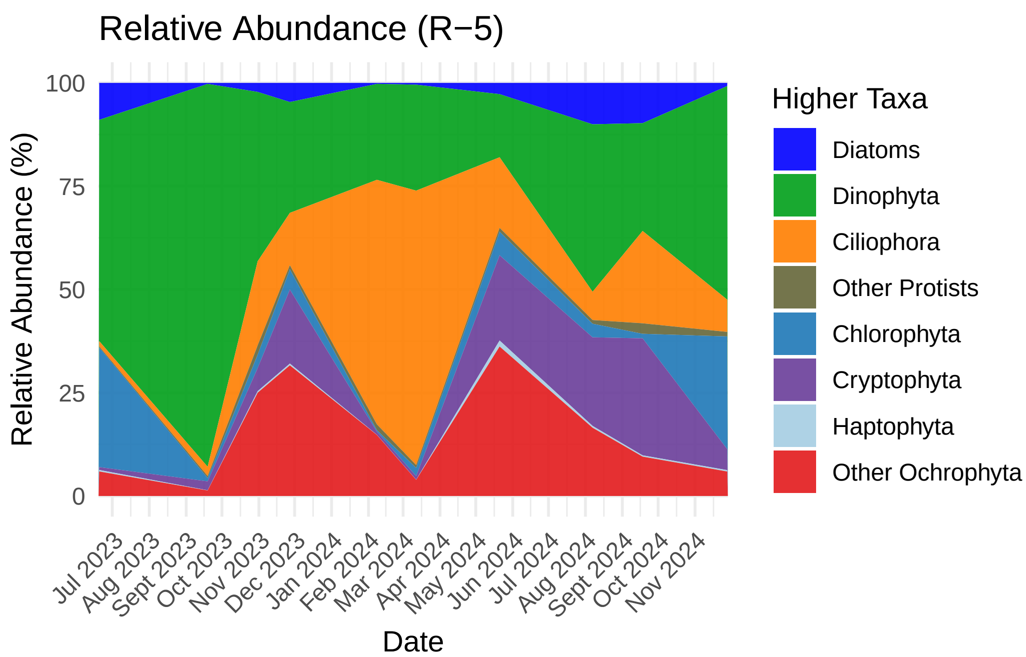

```r
library(patchwork)

library(patchwork)

library(patchwork)

(p_R5 + p_S9) +
  plot_layout(guides = "collect") +
  plot_annotation(tag_levels = "a") &
  theme(legend.position = "right")
```



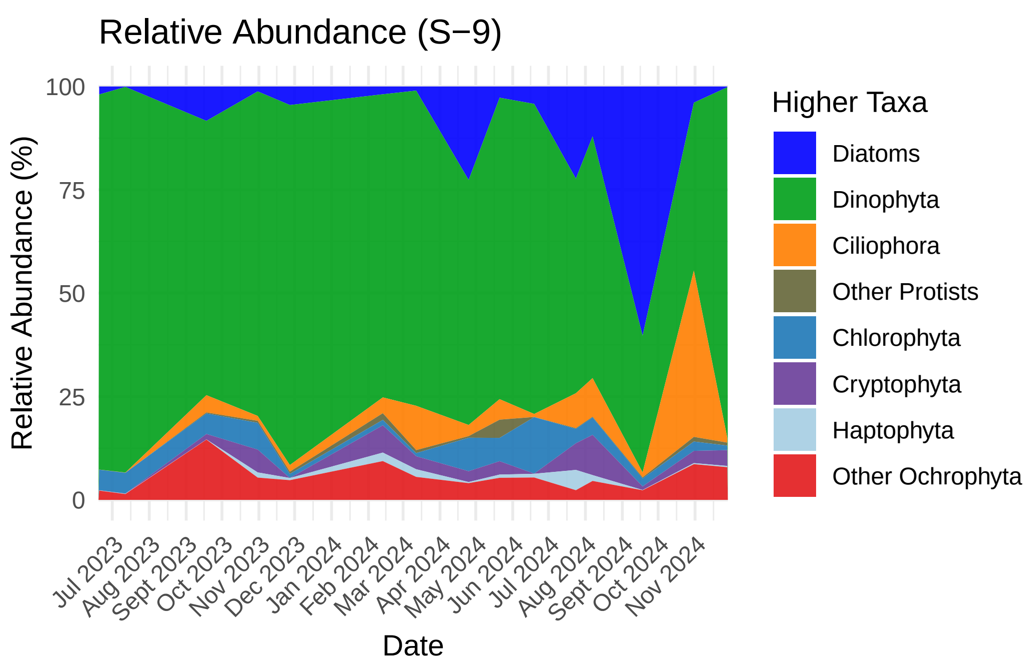
metabarcoding

```
# Your legend
all_taxa_merged <- c(
"Diatoms",
"Dinophyta",
"Ciliophora",
"Other Protists",
"Chlorophyta",
"Cryptophyta",
"Haptophyta",
"Euglenophyta",
"Other Ochrophyta"
)
# Ensure TAXON_COLORS has names matching all_taxa_merged; check and warn if not
missing_colors <- setdiff(all_taxa_merged, names(TAXON_COLORS_meta))
if (length(missing_colors) > 0) {
warning("TAXON_COLORS missing entries for: ", paste(missing_colors, collapse = ", "))
}
# Prepare data: types and factor ordering
data_long_meta <- data_long_meta %>%
```

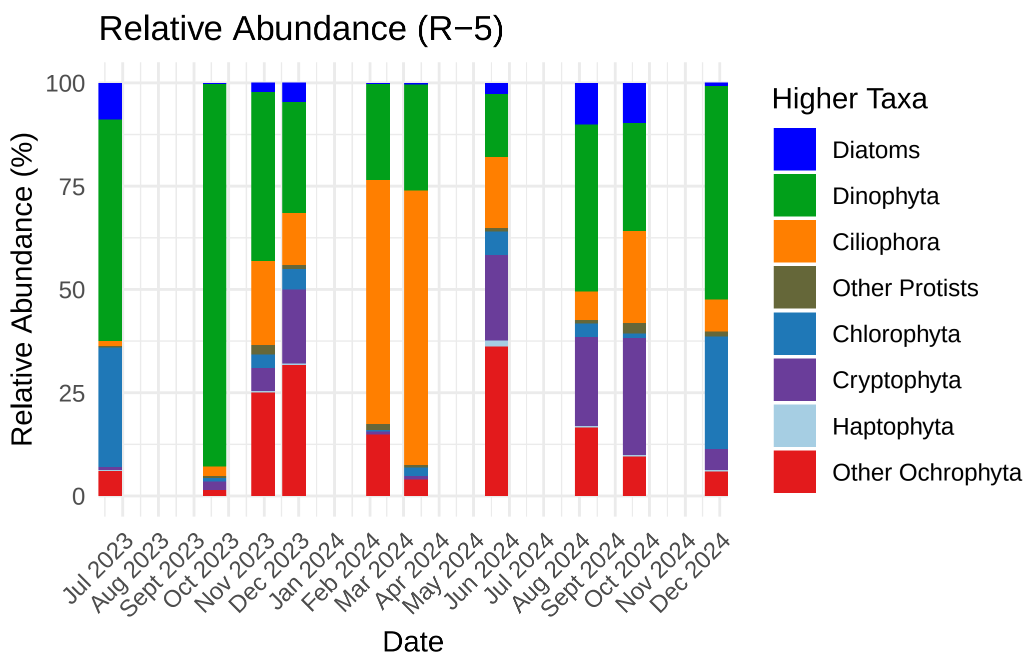
```r
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
# parse "yyyy-mm-dd"
higher_taxa = factor(higher_taxa, levels = all_taxa_merged)
) %>%
filter(!is.na(Date))
# Restrict to 2023-2024 and aggregate (sum) per day/station/taxon
df_prep_meta <- data_long_meta %>%
filter(Date >= as.Date("2023-01-01") & Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
  summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop")
# Optional: convert proportion to percent if needed
# df_prep_meta <- df_prep_meta %>% mutate(Abundance = Abundance * 100)
# Plot helper
make_area_plot_meta <- function(df, station_name) {
df_station <- df %>% filter(station == station_name)
ggplot(df_station, aes(x = Date, y = Abundance, fill = higher_taxa)) +
geom_area(position = "stack", color = NA, alpha = 0.9) +
scale_fill_manual(values = TAXON_COLORS_meta, na.value = "grey50") +
scale_x_date(date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
labs(
title = paste0("Relative Abundance (", station_name, ")"),
x = "Date",
y = "Relative Abundance (%)",
fill = "Higher Taxa"
) +
theme_minimal() +
theme(
axis.text.x = element_text(angle = 45, hjust = 1),
legend.position = "right"
)
}
# Create and show plots
area_meta_R5 <- make_area_plot_meta(df_prep_meta, "R-5")
area_meta_S9 <- make_area_plot_meta(df_prep_meta, "S-9")
print(area_meta_R5)
```




```
print(area_meta_S9)
```

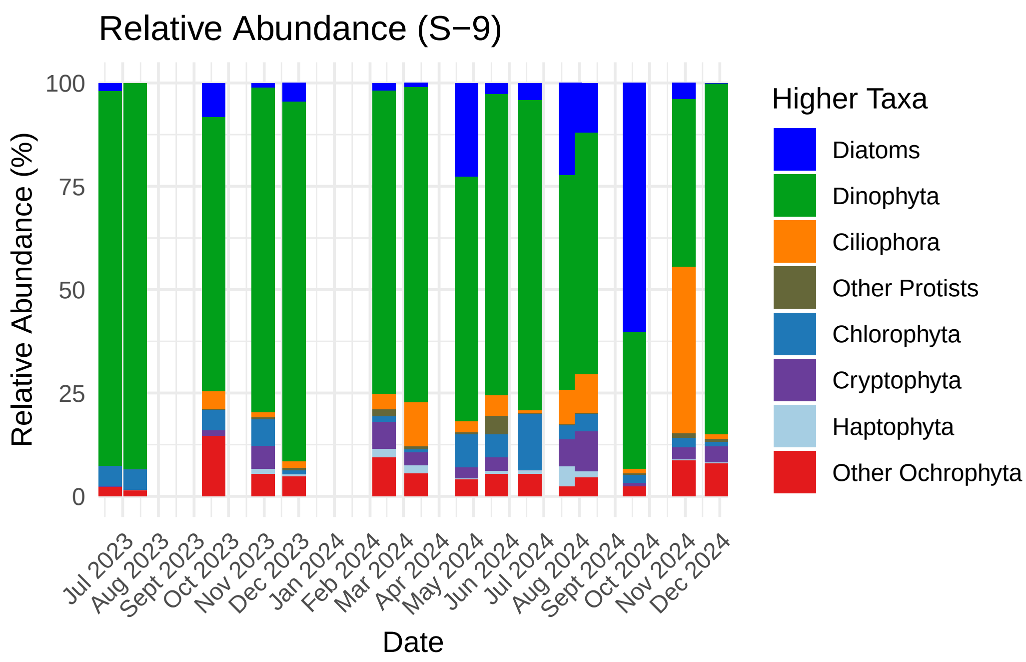


```
# Helper to build stacked bar (col) plot for a station
make_col_plot_meta <- function(df, station_name) {
df_station <- df %>% filter(station == station_name)
ggplot(df_station, aes(x = Date, y = Abundance, fill = higher_taxa)) +
geom_col(position = "stack", width = 20) + # width controls bar thickness; adjust as neede
scale_fill_manual(values = TAXON_COLORS_meta, na.value = "grey50") +
scale_x_date(date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
labs(
title = paste0("Relative Abundance (", station_name, ")"),
x = "Date",
y = "Relative Abundance (%)",
fill = "Higher Taxa"
) +
theme_minimal() +
theme(
axis.text.x = element_text(angle = 45, hjust = 1),
legend.position = "right"
)
}
# Create the bar plots
col_meta_R5 <- make_col_plot_meta(df_prep_meta, "R-5")
col_meta_S9 <- make_col_plot_meta(df_prep_meta, "S-9")
# Print them
print(col_meta_R5)
```




```
print(col_meta_S9)
```

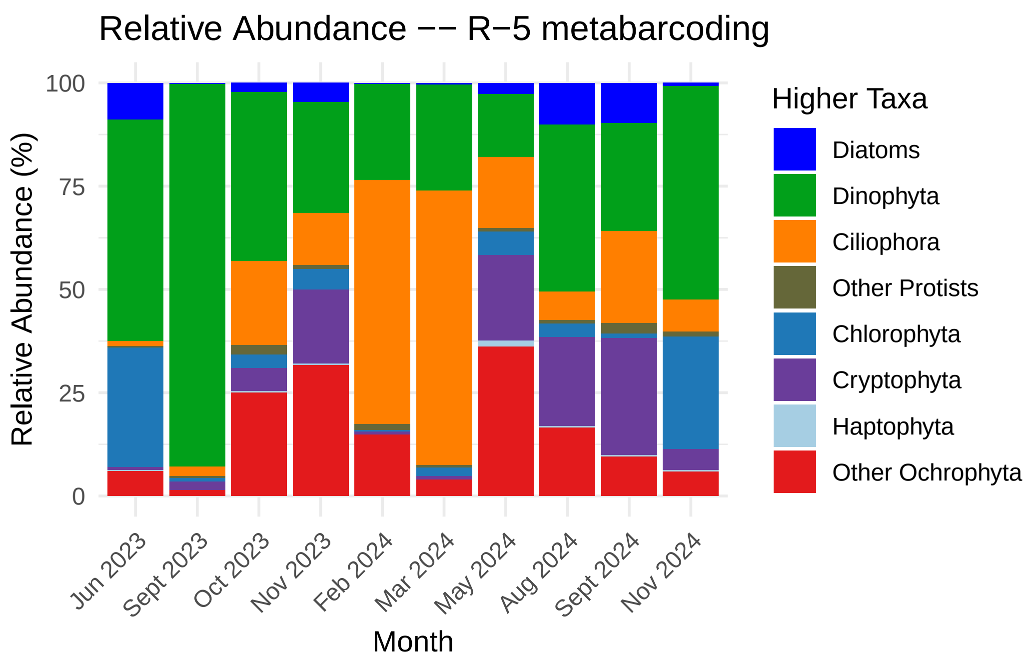
```
Warning: `position_stack()` requires non-overlapping x intervals.
```



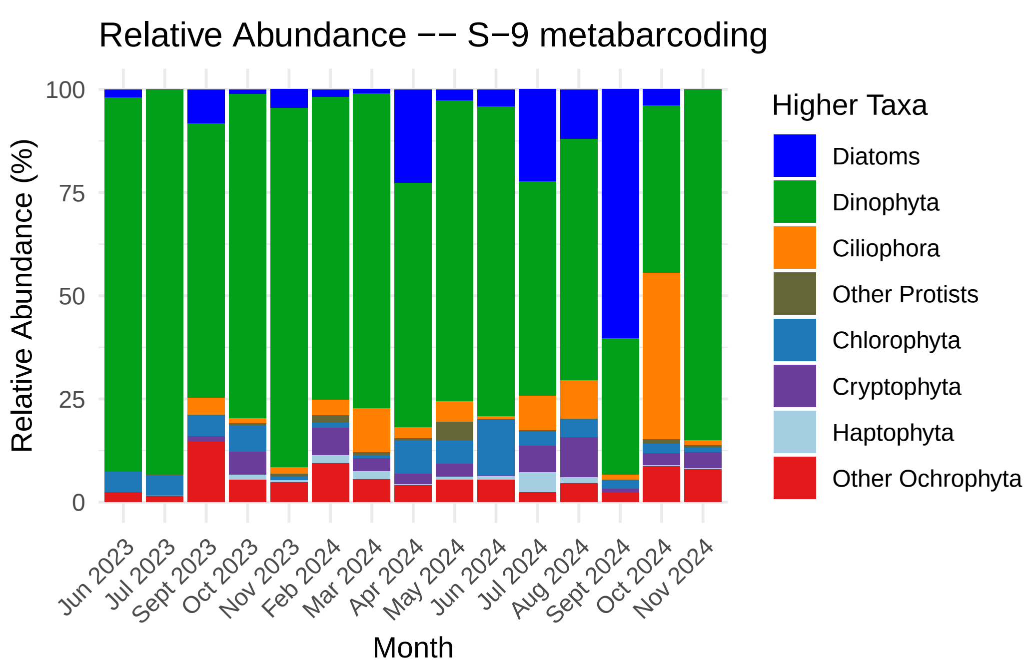
new plots

```
# Robust parsing orders (add formats you know if needed)
orders <- c("ymd","dmy","mdy", "Ymd", "dmy HMS", "mdy HMS", "dmy HM", "dmY", "Y/m/d", "d/m

# Prepare df_prep_meta for plotting: parse dates, drop NA Abundance, create month_label
df_meta_plot <- df_prep_meta %>%
  filter(!is.na(Abundance)) %>%
  mutate(
    raw_Date = as.character(Date),
    parsed_dt = suppressWarnings(parse_date_time(raw_Date, orders = orders, exact = FALSE)
    parsed_Date = as_date(parsed_dt),
    # month_label uses parsed date when available, otherwise the raw string
    month_label = if_else(!is.na(parsed_Date), format(parsed_Date, "%b %Y"), raw_Date),
    month_order_key = if_else(!is.na(parsed_Date), as.numeric(parsed_Date), NA_real_)
  )

# Build ordered factor levels: chronological parsed months first, then any unparsed labels
parsed_levels <- df_meta_plot %>%
  filter(!is.na(month_order_key)) %>%
  arrange(month_order_key) %>%
```

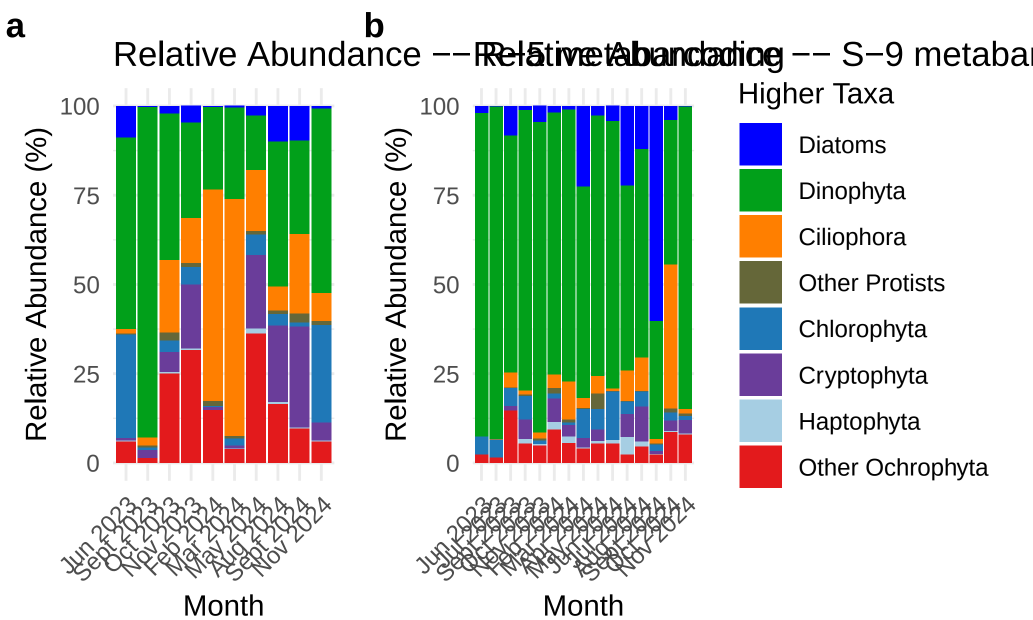
```
  pull(month_label) %>%
  unique()

unparsed_levels <- df_meta_plot %>%
  filter(is.na(month_order_key)) %>%
  pull(month_label) %>%
  unique()

all_levels <- c(parsed_levels, unparsed_levels)

df_meta_plot <- df_meta_plot %>%
  mutate(month_label = factor(month_label, levels = all_levels, ordered = TRUE))

# Helper to build stacked bar plot for a station (uses month_label)
make_col_plot_meta_fixed <- function(df, station_name, color_map = TAXON_COLORS_meta) {
  df %>%
    filter(station == station_name) %>%
    group_by(month_label, higher_taxa) %>%
    summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
    ggplot(aes(x = month_label, y = Abundance, fill = higher_taxa)) +
    geom_col(position = "stack", width = 0.9) +
    { if (!is.null(color_map)) scale_fill_manual(values = color_map, na.value = "grey50")
    labs(
      title = paste0("Relative Abundance -- ", station_name, " metabarcoding"),
      x = "Month",
      y = "Relative Abundance (%)",
      fill = "Higher Taxa"
    ) +
    theme_minimal() +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1),
      legend.position = "right"
    )
}

# Create the bar plots for the two stations
col_meta_R5 <- make_col_plot_meta_fixed(df_meta_plot, "R-5")
col_meta_S9 <- make_col_plot_meta_fixed(df_meta_plot, "S-9")

# Print or inspect
print(col_meta_R5)
```




```
print(col_meta_S9)
```



```r
(col_meta_R5 + col_meta_S9) +
  plot_layout(guides = "collect") +
  plot_annotation(tag_levels = "a") &
  theme(
    legend.position = "right",
    plot.tag = element_text(face = "bold")
  )
```

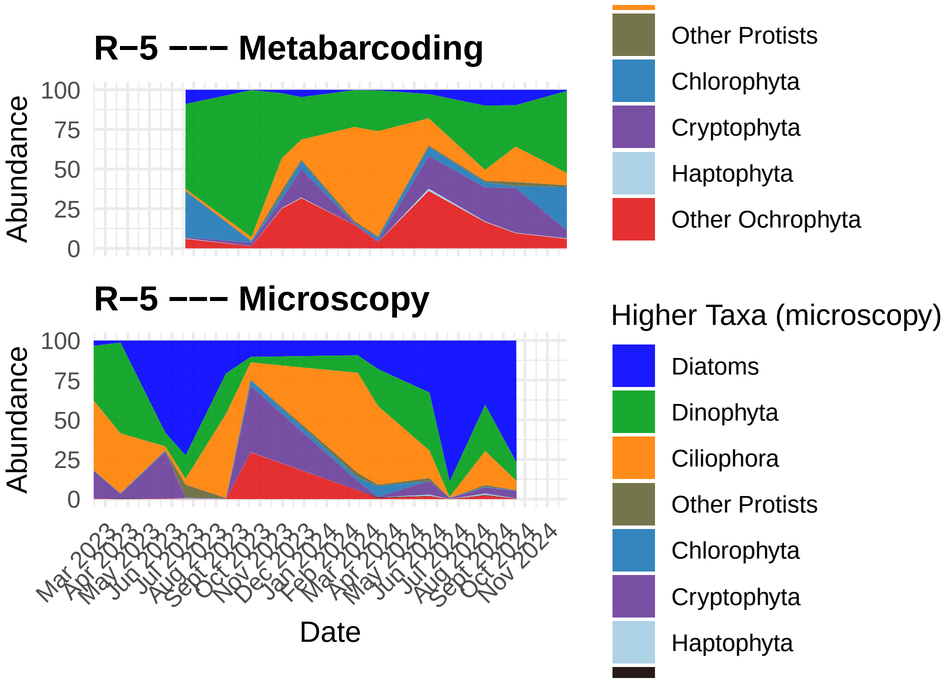


Combined

```r
df_mic <- combined_long_microscopy %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
  summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
```

```
filter(station == "R-5")
# Metabarcoding / meta dataset (data_long_meta)
df_meta <- data_long_meta %>%
  mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "R-5")


# Determine a common date range to ensure x-axis alignment
date_min <- min(c(min(df_mic$Date, na.rm = TRUE), min(df_meta$Date, na.rm = TRUE)), na.rm
date_max <- max(c(max(df_mic$Date, na.rm = TRUE), max(df_meta$Date, na.rm = TRUE)), na.rm

# Metabarcoding (top)
area_meta <- ggplot(df_meta, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_area(position = "stack", color = NA, alpha = 0.9) +
  scale_fill_manual(
    name = "Higher Taxa (metabarcoding)",
    values = TAXON_COLORS_meta,
    breaks = all_taxa_merged
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "R-5 --- Metabarcoding",
       x = NULL, y = "Abundance") +   # hide x label on top plot
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),   # hide x labels on top plot
    axis.ticks.x = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# Microscopy (bottom)
area_mic <- ggplot(df_mic, aes(x = Date, y = Abundance, fill = higher_taxa)) +
```

```r
  geom_area(position = "stack", color = NA, alpha = 0.9) +
  scale_fill_manual(
    name = "Higher Taxa (microscopy)",
    values = TAXON_COLORS_micro,
    breaks = all_taxa_merged_micro
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "R-5 --- Microscopy",
       x = "Date", y = "Abundance") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# Combine vertically: metabarcoding (top) / microscopy (bottom)
combined_plot <- area_meta / area_mic +
  plot_layout(heights = c(1, 1), guides = "collect") &
  theme(legend.position = "right")   # collect shared legend on the right

print(combined_plot)
```



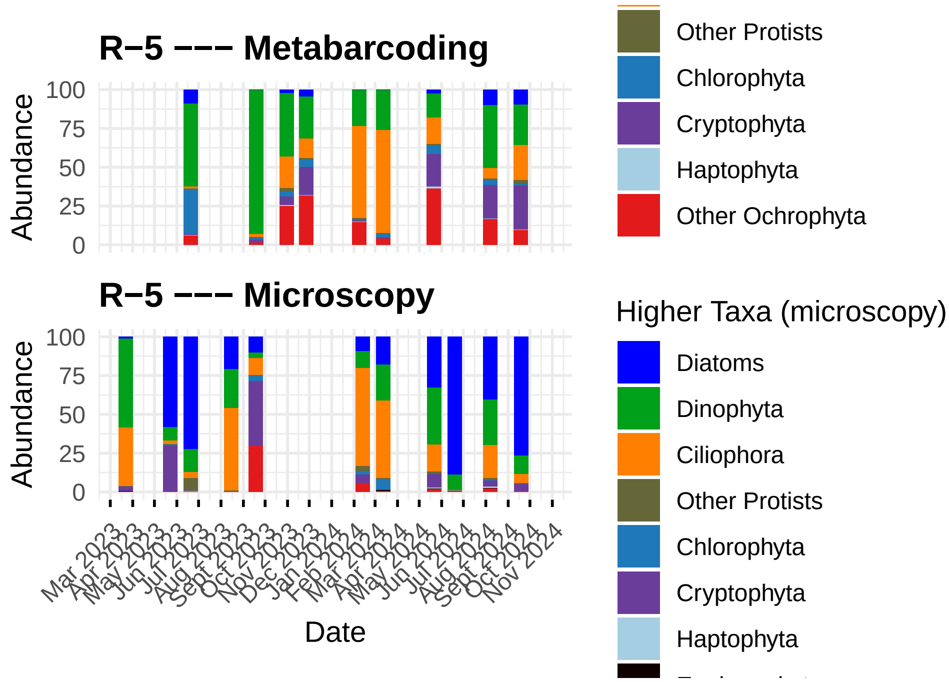

```
df_mic <- combined_long_microscopy %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "R-5")
#--- Prepare metabarcoding/meta dataframe (data_long_meta)--
df_meta <- data_long_meta %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
```

```r
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "R-5")
# optional checks
if (nrow(df_mic) == 0) message("No microscopy rows for R-5 in the date range.")
if (nrow(df_meta) == 0) message("No metabarcoding rows for R-5 in the date range.")
# common date limits so x axes align
```

```r
date_min <- min(c(min(df_mic$Date, na.rm = TRUE), min(df_meta$Date, na.rm = TRUE)), na.rm
date_max <- max(c(max(df_mic$Date, na.rm = TRUE), max(df_meta$Date, na.rm = TRUE)), na.rm


# width for geom_col (in days)- adjust to sampling frequency
col_width <- 20

# --- Microscopy stacked column plot (bottom) ---
col_mic <- ggplot(df_mic, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_col(position = "stack", width = col_width) +
  scale_fill_manual(
    name = "Higher Taxa (microscopy)",
    values = TAXON_COLORS_micro,
    breaks = all_taxa_merged_micro
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "R-5 --- Microscopy", x = "Date", y = "Abundance") +  # show x label here
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),   # SHOW x labels on bottom
    axis.ticks.x = element_line(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Metabarcoding stacked column plot (top) ---
col_meta <- ggplot(df_meta, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_col(position = "stack", width = col_width) +
  scale_fill_manual(
```

```r
    name = "Higher Taxa (metabarcoding)",
    values = TAXON_COLORS_meta,
    breaks = all_taxa_merged
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "R-5 --- Metabarcoding", x = NULL, y = "Abundance") +  # hide x label on to
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),   # HIDE x labels on top
    axis.ticks.x = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Stack vertically: metabarcoding (top) / microscopy (bottom) ---
# Use guides = "collect" if you want a single combined legend
combined_cols <- col_meta / col_mic + plot_layout(heights = c(1, 1), guides = "collect") &
  theme(legend.position = "right")

print(combined_cols)
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_col()`).

Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_col()`).
```



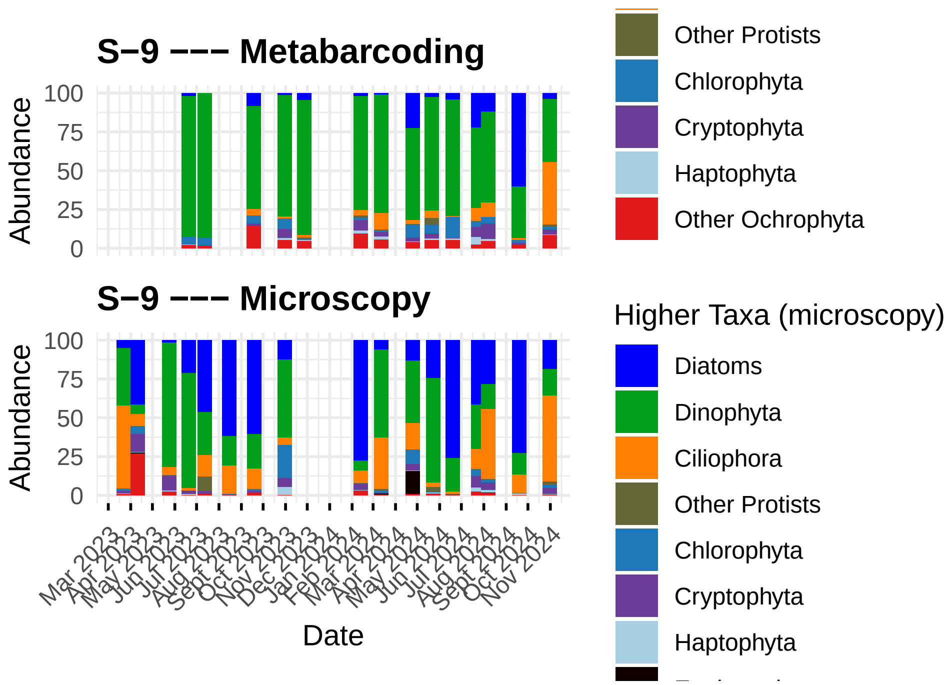

```
#--- Prepare microscopy dataframe (combined_long_microscopy) for S-9--
df_mic_S9 <- combined_long_microscopy %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged_micro)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "S-9")


#--- Prepare metabarcoding/meta dataframe (data_long_meta) for S-9--
df_meta_S9 <- data_long_meta %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged)
) %>%
```

```r
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "S-9")


# common date limits so x axes align
date_min <- min(c(min(df_mic_S9$Date, na.rm = TRUE),
                  min(df_meta_S9$Date, na.rm = TRUE)), na.rm = TRUE)
date_max <- max(c(max(df_mic_S9$Date, na.rm = TRUE),
                  max(df_meta_S9$Date, na.rm = TRUE)), na.rm = TRUE)

# width for geom_col (in days) - adjust to sampling frequency
col_width <- 20

# --- Metabarcoding stacked column plot for S-9 (TOP) ---
col_meta_S9 <- ggplot(df_meta_S9, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_col(position = "stack", width = col_width) +
  scale_fill_manual(
    name = "Higher Taxa (metabarcoding)",
    values = TAXON_COLORS_meta,
    breaks = all_taxa_merged
  ) +
  scale_x_date(limits = c(date_min, date_max),
             date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "S-9 --- Metabarcoding", x = NULL, y = "Abundance") +   # hide x label on t
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),   # HIDE x labels on top
    axis.ticks.x = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Microscopy stacked column plot for S-9 (bottom) ---
col_mic_S9 <- ggplot(df_mic_S9, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_col(position = "stack", width = col_width) +
  scale_fill_manual(
    name = "Higher Taxa (microscopy)",
    values = TAXON_COLORS_micro,
    breaks = all_taxa_merged_micro
```

```r
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "S-9 --- Microscopy", x = "Date", y = "Abundance") +  # show x label here
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),   # SHOW x labels on bottom
    axis.ticks.x = element_line(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Metabarcoding stacked column plot for S-9 (top) ---
col_meta_S9 <- ggplot(df_meta_S9, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_col(position = "stack", width = col_width) +
  scale_fill_manual(
    name = "Higher Taxa (metabarcoding)",
    values = TAXON_COLORS_meta,
    breaks = all_taxa_merged
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "S-9 --- Metabarcoding", x = NULL, y = "Abundance") +  # hide x label on to
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),   # HIDE x labels on top
    axis.ticks.x = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Stack vertically: metabarcoding (top) / microscopy (bottom) ---
combined_cols_S9 <- col_meta_S9 / col_mic_S9 +
  plot_layout(heights = c(1, 1), guides = "collect") &
  theme(legend.position = "right")

# show
print(combined_cols_S9)
```

```
Warning: `position_stack()` requires non-overlapping x intervals.

Warning: Removed 8 rows containing missing values or values outside the scale range
```

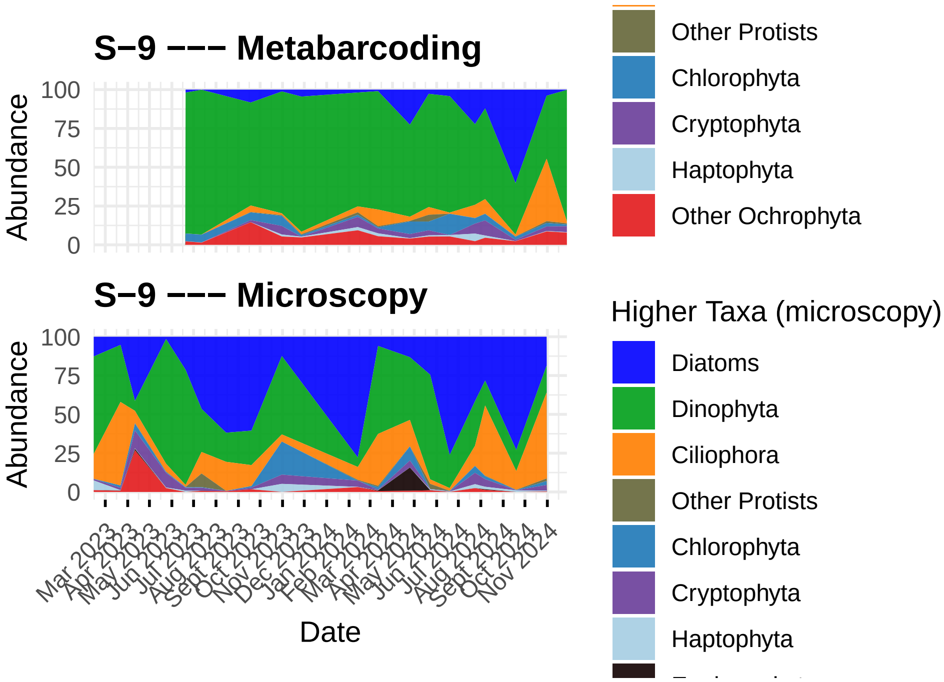
```
(`geom_col()`).

Warning: `position_stack()` requires non-overlapping x intervals.

Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_col()`).
```




```r
#--- Prepare microscopydataframe(combined_long_microscopy)forS-9--
df_mic_S9<-combined_long_microscopy %>%
mutate(
Date= as.character(Date),
Date= ymd(Date),
Abundance= as.numeric(Abundance),
higher_taxa= factor(higher_taxa, levels= all_taxa_merged_micro)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "S-9")
```

```
#--- Prepare metabarcoding/meta dataframe (data_long_meta) for S-9--
df_meta_S9 <- data_long_meta %>%
mutate(
Date = as.character(Date),
Date = ymd(Date),
Abundance = as.numeric(Abundance),
higher_taxa = factor(higher_taxa, levels = all_taxa_merged)
) %>%
filter(!is.na(Date), Date >= as.Date("2023-01-01"), Date <= as.Date("2024-12-31")) %>%
group_by(station, Date, higher_taxa) %>%
summarise(Abundance = sum(Abundance, na.rm = TRUE), .groups = "drop") %>%
filter(station == "S-9")


# Determine common x-axis limits so the two panels align
date_min <- min(
  c(min(df_mic_S9$Date, na.rm = TRUE),
    min(df_meta_S9$Date, na.rm = TRUE)),
  na.rm = TRUE
)
date_max <- max(
  c(max(df_mic_S9$Date, na.rm = TRUE),
    max(df_meta_S9$Date, na.rm = TRUE)),
  na.rm = TRUE
)

#--- Build microscopy area plot (uses TAXON_COLORS_micro and all_taxa_merged_micro)--
# --- microscopy area plot (S-9) - BOTTOM panel ---
area_mic_S9 <- ggplot(df_mic_S9, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_area(position = "stack", color = NA, alpha = 0.9) +
  scale_fill_manual(
    name = "Higher Taxa (microscopy)",
    values = TAXON_COLORS_micro,
    breaks = all_taxa_merged_micro
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "S-9 --- Microscopy", x = "Date", y = "Abundance") +   # show x label here
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),  # SHOW x labels on bottom panel
```

```
    axis.ticks.x = element_line(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Build metabarcoding area plot (S-9) - TOP panel ---
area_meta_S9 <- ggplot(df_meta_S9, aes(x = Date, y = Abundance, fill = higher_taxa)) +
  geom_area(position = "stack", color = NA, alpha = 0.9) +
  scale_fill_manual(
    name = "Higher Taxa (metabarcoding)",
    values = TAXON_COLORS_meta,
    breaks = all_taxa_merged
  ) +
  scale_x_date(limits = c(date_min, date_max),
               date_breaks = "1 month", date_labels = "%b %Y", expand = c(0, 0)) +
  labs(title = "S-9 --- Metabarcoding", x = NULL, y = "Abundance") +  # hide x label on to
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),   # HIDE x labels on top panel
    axis.ticks.x = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

# --- Combine vertically: metabarcoding (top) / microscopy (bottom) ---
combined_area_S9 <- area_meta_S9 / area_mic_S9 +
  plot_layout(heights = c(1, 1), guides = "collect") &   # use guides="collect" to merge l
  theme(legend.position = "right")

# Print
print(combined_area_S9)
```



Sandras code

combining datasets

```
#combining df_prep and df_prep_meta
# df_prep: rename to microscopy_abundance (overwrite)
if ("Abundance" %in% names(df_prep)) {
df_prep <- df_prep %>%
rename(microscopy_abundance = Abundance)
} else {
warning("df_prep does not contain a column named 'abundance'")
}

# df_prep_meta: rename to metabarcoding_abundance (overwrite)
if ("Abundance" %in% names(df_prep_meta)) {
df_prep_meta <- df_prep_meta %>%
  rename(metabarcoding_abundance = Abundance)
} else {
warning("df_prep_meta does not contain a column named 'abundance'")
}
```

```
# Replace if your higher-taxa column has a different name:
key_station <- "station"
key_date<- "Date"
key_taxa<- "higher_taxa" # <- change this to your real column name

# Ensure date columns are Date type (ISO yyyy-mm-dd works with as.Date)
df_prep[[key_date]]<- as.Date(df_prep[[key_date]])
df_prep_meta[[key_date]] <- as.Date(df_prep_meta[[key_date]])

# Change the summary function if you prefer mean or first instead of sum.
df_prep_meta_agg <- df_prep_meta %>%
group_by(across(all_of(c(key_station, key_date, key_taxa)))) %>%
summarise(metabarcoding_abundance = sum(metabarcoding_abundance, na.rm = TRUE),
.groups = "drop")

# Left join the metabarcoding abundance into the microscopy dataframe
Combined_micro_meta <- df_prep %>%
left_join(df_prep_meta_agg, by = c(key_station, key_date, key_taxa))

#split the datasett into stationed datasets
Combined_micro_meta_S_9 <- Combined_micro_meta %>%
filter(station == "S-9") %>%
select(-station)
Combined_micro_meta_R_5 <- Combined_micro_meta %>%
filter(station == "R-5") %>%
select(-station)
```

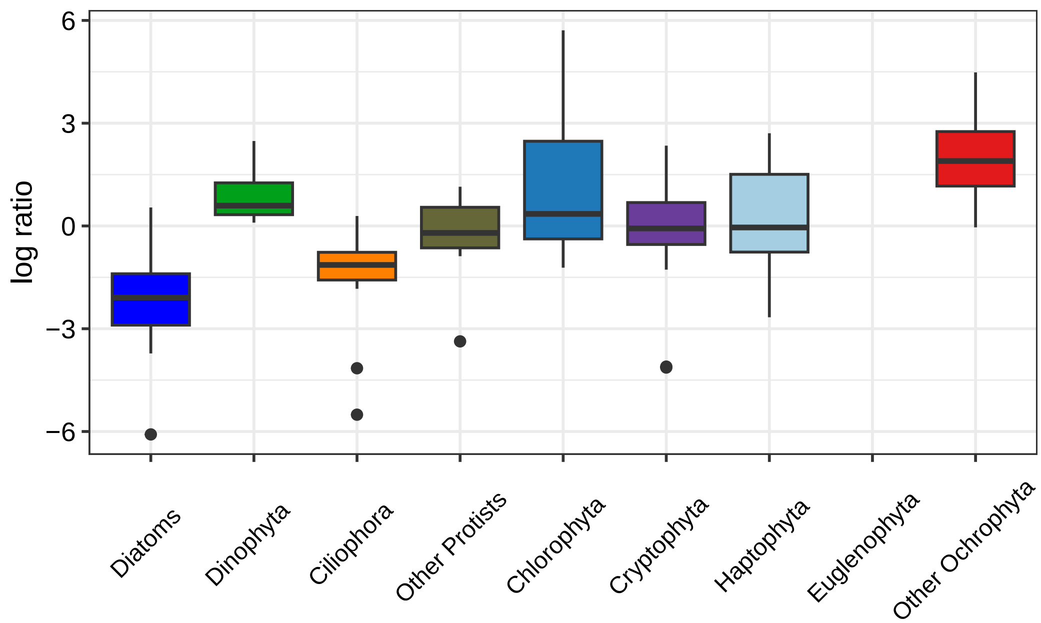
S-9

```
# 1) proper month order (no duplicates, include JUL)
month_order <- c(
  "JAN.23", "FEB.23", "MAR.23", "APR.23", "MAY.23", "JUN.23", "JUL.23",
  "AUG.23", "SEP.23", "OCT.23", "NOV.23", "DEC.23",
  "JAN.24", "FEB.24", "MAR.24", "APR.24", "MAY.24", "JUN.24", "JUL.24",
  "AUG.24", "SEP.24", "OCT.24", "NOV.24", "DEC.24"
)

# 2) prepare data, create month labels, handle zeros/NA before log
count_reads <- Combined_micro_meta_S_9 %>%
  mutate(
    # ensure Date is Date class
```

```
    Date = as.Date(Date),
    # create label like "JAN.23"
    month = toupper(format(Date, "%b.%y")),   # e.g. "Jan.23" -> "JAN.23"
    month = factor(month, levels = month_order),
    # optional: replace NA with 0 or leave as NA depending on your logic
    # handle zeros: choose one strategy below

    # Strategy A: filter out rows where microscopy_abundance is zero or NA
    # (uncomment if you prefer this)
    # microscopy_abundance = as.numeric(microscopy_abundance),
    # metabarcoding_abundance = as.numeric(metabarcoding_abundance)

    # Strategy B (use small pseudocount to avoid Inf on division & log)
    microscopy_abundance = as.numeric(microscopy_abundance),
    metabarcoding_abundance = as.numeric(metabarcoding_abundance),
    pseudocount = 1e-6,
    microscopy_adj = if_else(is.na(microscopy_abundance), NA_real_, microscopy_abundance +
    metabarcoding_adj = if_else(is.na(metabarcoding_abundance), NA_real_, metabarcoding_ab

    ratio = metabarcoding_adj / microscopy_adj,
    log_ratio = if_else(is.finite(ratio) & ratio > 0, log(ratio), NA_real_)
  ) %>%
  select(-pseudocount, -microscopy_adj, -metabarcoding_adj)




# Log-ratios
count_reads <- Combined_micro_meta %>%
  mutate(
    month = factor(Date, levels = month_order),
    ratio = metabarcoding_abundance / microscopy_abundance,
    log_ratio = log(ratio)
  )


readbiov_plot <- ggplot(count_reads, aes(x = higher_taxa, y = log_ratio, fill = higher_tax
  geom_boxplot() +
  scale_fill_manual(values = TAXON_COLORS_micro, na.value = "red") +
  theme_bw() +
  theme(
```

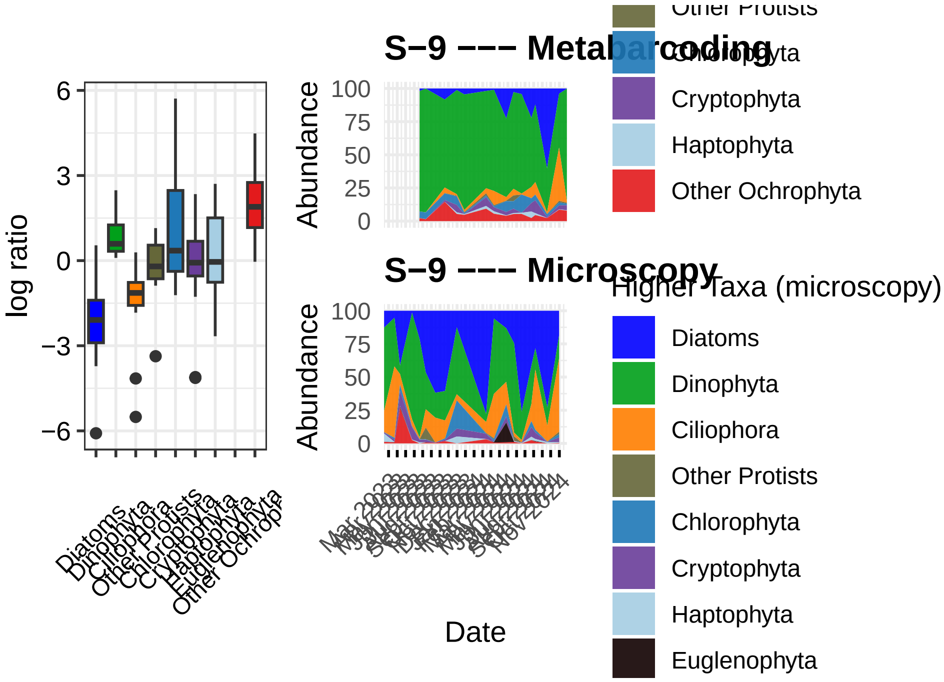
```
    axis.text.x   = element_text(color = "black", size = 9, angle = 45, vjust = 0.55, hjus
    axis.text.y   = element_text(color = "black", size = 10),
    legend.title  = element_text(colour = "white", size = 14, face = "bold"),
    legend.text   = element_text(colour = "black", size = 10),
    legend.position = "none"
  ) +
  labs(x = "", y = "log ratio")


# Time series for reads vs biovolumes
count_reads_long <- count_reads %>%
  pivot_longer(
    cols = c(metabarcoding_abundance, microscopy_abundance),
    names_to = "type",
    values_to = "value"
  ) %>%
  mutate(
    type = factor(type, levels = c("metabarcoding_abundance", "microscopy_abundance"))
  )

readbiov_season <- ggplot(count_reads_long, aes(x = Date, y = value, fill = higher_taxa, h
  facet_grid(type ~ ., scales = "free_y") +
  geom_area(position = "fill") +    scale_fill_manual(values = TAXON_COLORS_micro, na.valu
  theme_bw() +
  theme(
    axis.text.x  = element_text(color = "black", size = 9, angle = 45, vjust = 0.7),
    axis.text.y  = element_text(color = "black", size = 9),
    axis.title.x = element_text(face = "bold", colour = "black", size = 9, vjust = -5),
    axis.title.y = element_text(colour = "black", size = 11, vjust = 1, hjust = 0.54),
    legend.title = element_text(colour = "white", size = 9, face = "bold"),
    legend.text  = element_text(colour = "black", size = 8)
  ) +
  labs(x = "", y = "")

readbiov_plot
```

```
Warning: Removed 174 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




```
combined_plot_S_9 <- readbiov_plot + combined_area_S9
combined_plot_S_9
```

```
Warning: Removed 174 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



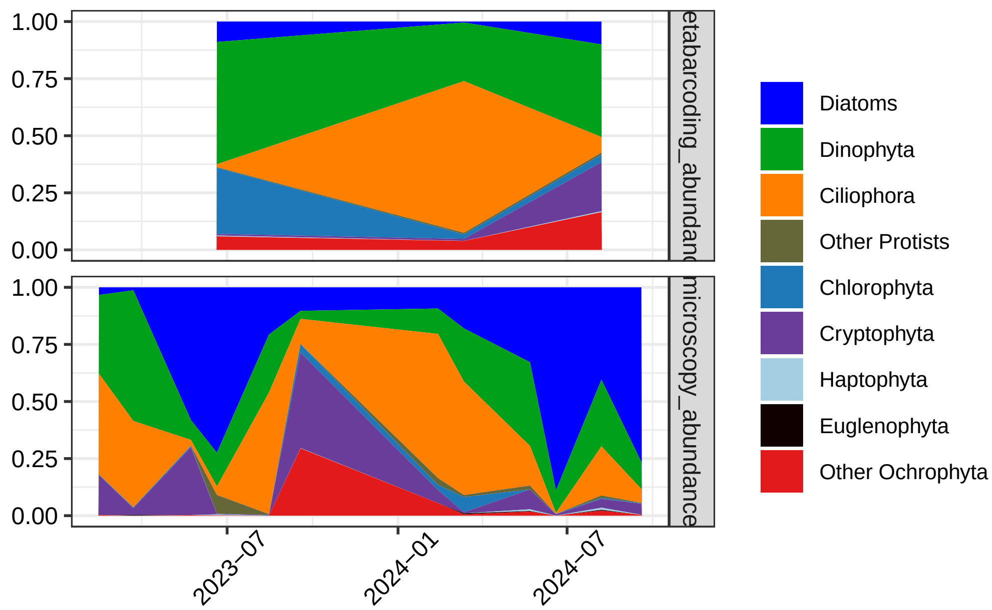
R-5

```
# 2) prepare data, create month labels, handle zeros/NA before log
count_reads_R_5 <- Combined_micro_meta_R_5 %>%
  mutate(
    # ensure Date is Date class
    Date = as.Date(Date),
    # create label like "JAN.23"
    month = toupper(format(Date, "%b.%y")),   # e.g. "Jan.23" -> "JAN.23"
    month = factor(month, levels = month_order),
    # optional: replace NA with 0 or leave as NA depending on your logic
    # handle zeros: choose one strategy below

    # Strategy A: filter out rows where microscopy_abundance is zero or NA
    # (uncomment if you prefer this)
    # microscopy_abundance = as.numeric(microscopy_abundance),
    # metabarcoding_abundance = as.numeric(metabarcoding_abundance)

    # Strategy B (use small pseudocount to avoid Inf on division & log)
    microscopy_abundance = as.numeric(microscopy_abundance),
    metabarcoding_abundance = as.numeric(metabarcoding_abundance),
```

```r
    pseudocount = 1e-6,
    microscopy_adj = if_else(is.na(microscopy_abundance), NA_real_, microscopy_abundance +
    metabarcoding_adj = if_else(is.na(metabarcoding_abundance), NA_real_, metabarcoding_ab

    ratio = metabarcoding_adj / microscopy_adj,
    log_ratio = if_else(is.finite(ratio) & ratio > 0, log(ratio), NA_real_)
  ) %>%
  select(-pseudocount, -microscopy_adj, -metabarcoding_adj)



readbiov_plot_R_5 <- ggplot(count_reads_R_5, aes(x = higher_taxa, y = log_ratio, fill = hi
  geom_boxplot() +
  scale_fill_manual(values = TAXON_COLORS_micro, na.value = "red") +
  theme_bw() +
  theme(
    axis.text.x   = element_text(color = "black", size = 9, angle = 45, vjust = 0.55, hjus
    axis.text.y   = element_text(color = "black", size = 10),
    legend.title  = element_text(colour = "white", size = 14, face = "bold"),
    legend.text   = element_text(colour = "black", size = 10),
    legend.position = "none"
  ) +
  labs(x = "", y = "log ratio")


# Time series for reads vs biovolumes
count_reads_long_R_5 <- count_reads_R_5 %>%
  pivot_longer(
    cols = c(metabarcoding_abundance, microscopy_abundance),
    names_to = "type",
    values_to = "value"
  ) %>%
  mutate(
    type = factor(type, levels = c("metabarcoding_abundance", "microscopy_abundance"))
  )

readbiov_season_R_5 <- ggplot(count_reads_long_R_5, aes(x = Date, y = value, fill = higher
  facet_grid(type ~ ., scales = "free_y") +
  geom_area(position = "fill") +    scale_fill_manual(values = TAXON_COLORS_micro, na.valu
  theme_bw() +
  theme(
```

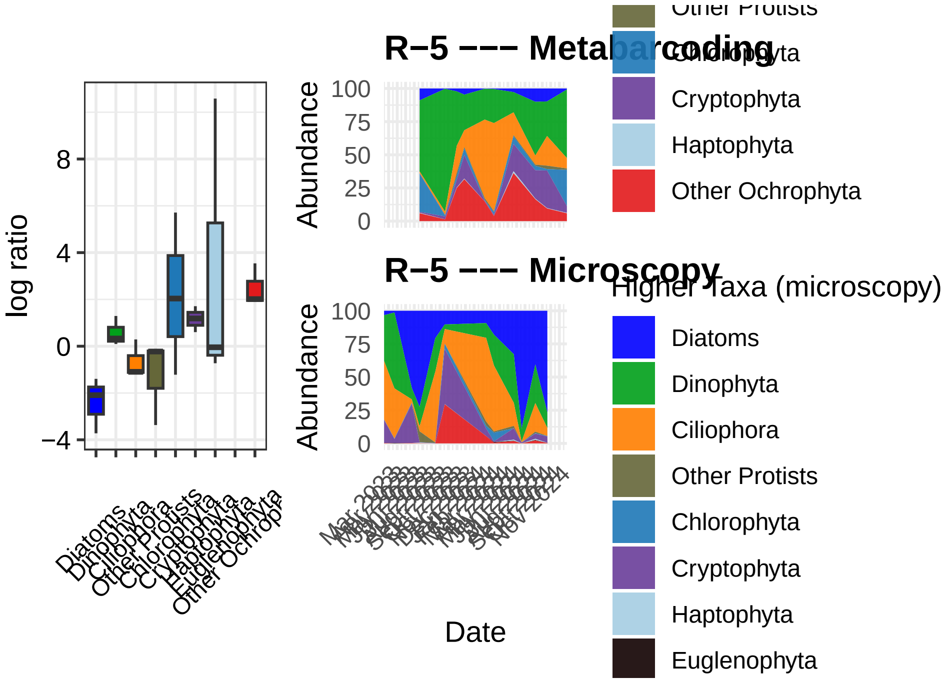
```
    axis.text.x  = element_text(color = "black", size = 9, angle = 45, vjust = 0.7),
    axis.text.y  = element_text(color = "black", size = 9),
    axis.title.x = element_text(face = "bold", colour = "black", size = 9, vjust = -5),
    axis.title.y = element_text(colour = "black", size = 11, vjust = 1, hjust = 0.54),
    legend.title = element_text(colour = "white", size = 9, face = "bold"),
    legend.text  = element_text(colour = "black", size = 8)
  ) +
  labs(x = "", y = "")

readbiov_season_R_5
```

```
Warning: Removed 84 rows containing non-finite outside the scale range
(`stat_align()`).
```




```
combined_plot_R_5 <- readbiov_plot_R_5 + combined_plot
combined_plot_R_5
```

```
Warning: Removed 84 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



R−5 −−− Metabarcoding
R−5 −−− Microscopy
log ratio
Abundance
Abundance
Date
Other Protists
Chlorophyta
Cryptophyta
Haptophyta
Other Ochrophyta
Higher Taxa (microscopy)
Diatoms
Dinophyta
Ciliophora
Other Protists
Chlorophyta
Cryptophyta
Haptophyta
Euglenophyta